



HOW TO EXPLORE YOUR DATA

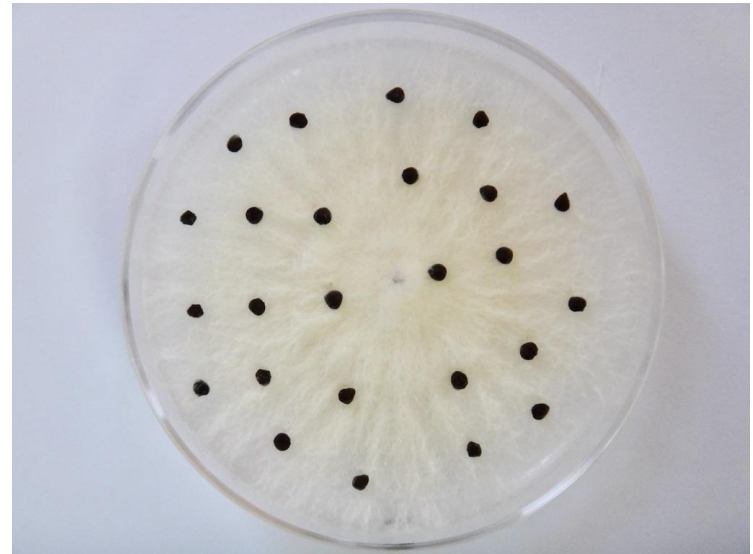
With EXCEL

Carolina Sarmiento
USF
October 9, 2026

A little bit about me



- Microbial ecologist (tropical)
- Plant-fungal interactions
- Working mostly on seeds
(but also, some roots, leaves)
- I work with:



Dr. Luanna Prevost – Amgen Biotech Experience (Master Lab Tech – Tampa Site Coordinator)
Dr. Camilo Zalamea – Lab Manager/Research
Smithsonian Tropical Research Institute – Research Associate

Writing and Reviewing



Publications in:

- ☐ Annual Review of Ecology, Evolution, and Systematics (AREES)
- ☐ PNAS
- ☐ Science
 - ☐ Proceedings of the Royal Society B
- ☐ New Phytologist
- ☐ Ecology
 - ☐ Functional Ecology
- ☐ Biotropica
- ☐ Others

Reviewer for:

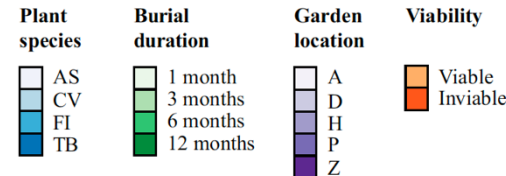
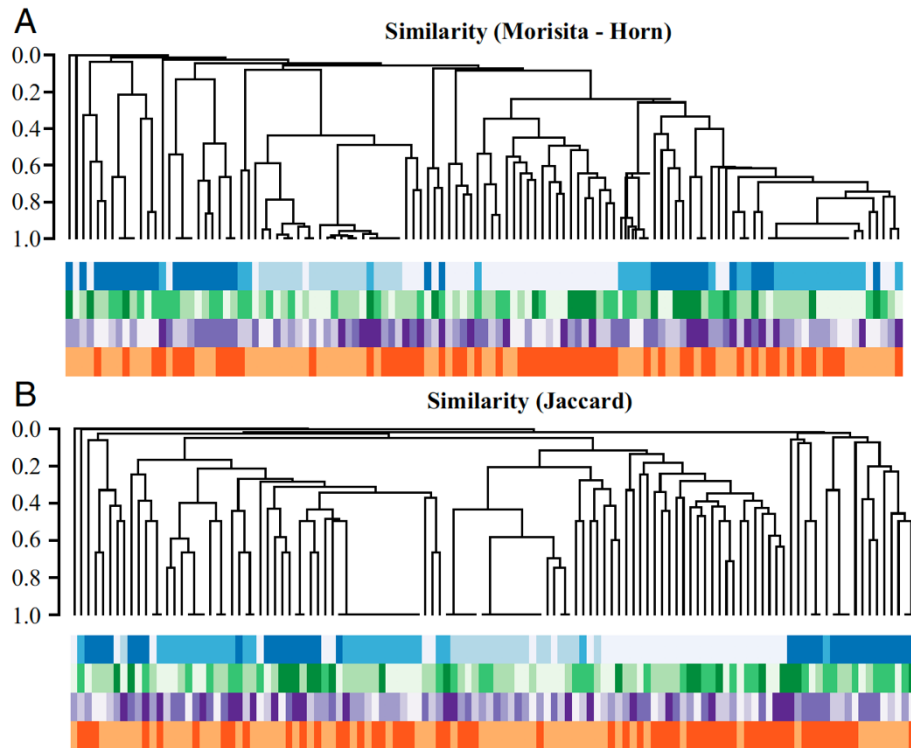
- ☐ New Phytologist
- ☐ Functional Ecology
- ☐ Fungal Ecology
- ☐ Ecology and Evolution
- ☐ Biotropica
- ☐ Journal of Ecology

[My profile on Scholar Google](#)

I enjoy ...



Lab work and data analysis and visualization!



```
pdf(file="Figures/cluster_9spp_Jaccard_revision_19jun2017.pdf", paper = "special", width
6, height = 8, family="Times")
layout(matrix(c(1,2,3), 1, 3, byrow = TRUE)) #divide your plotting device in 4
# draw the dendrogram == from above
par(cex=0.7, mai=c(.4,.3,.2,.1))
plot(as.dendrogram(ca), horiz= TRUE, leaflab="none")
title(ylab="Similarity (Jaccard)")
#mtext("ALL SEEDS - 9spp", cex = .7)

# draw your heatmap
#mtext("ALL SEEDS - 9spp", cex = .7)
#mtext("ALL SEEDS - 9spp", cex = .7)
```

I enjoy ...



Data analysis and data visualization!

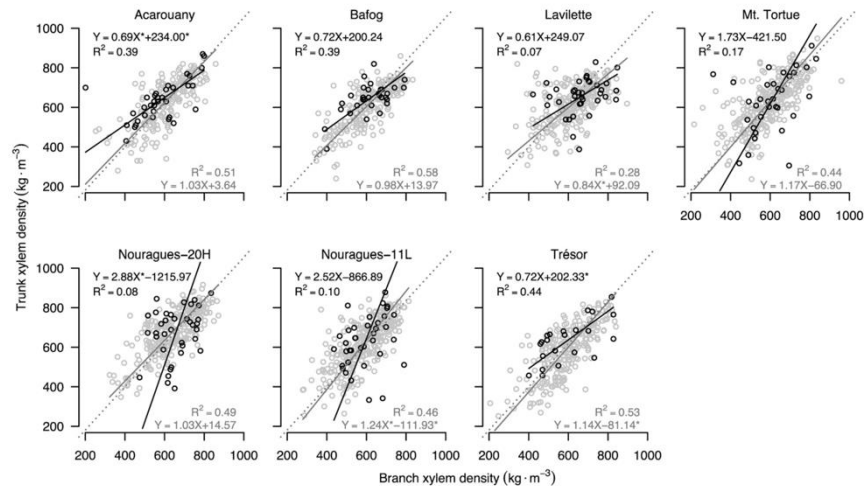


Fig. 3. The major-axis (MA) fit between trunk and branch xylem density in seven rain forest sites from French Guiana. The gray points and solid gray line indicate the site-level relationship, whereas the black points and lines indicate individuals of Fabaceae in each site. The dotted line represents the MA-fitted line for all 1909 samples together. Asterisks indicate significance as in Fig. 1.

I enjoy ...



Data analysis and data visualization!

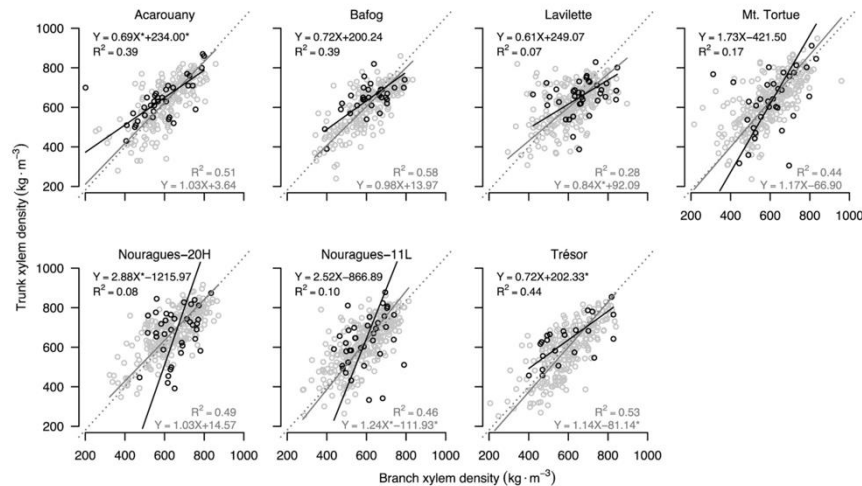
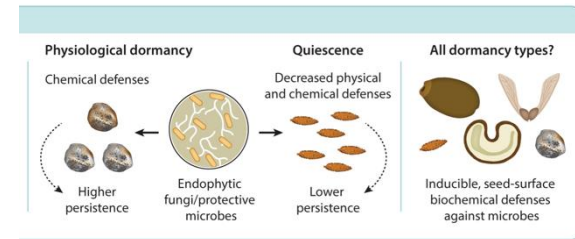
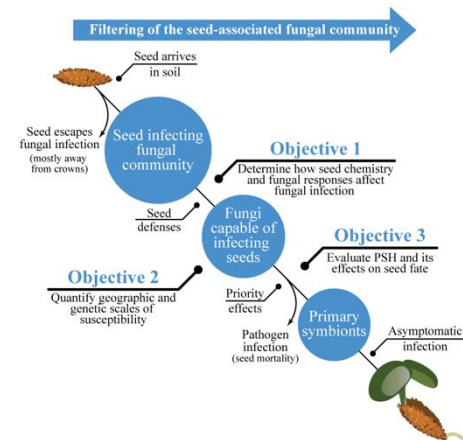


Fig. 3. The major-axis (MA) fit between trunk and branch xylem density in seven rain forest sites from French Guiana. The gray points and solid gray line indicate the site-level relationship, whereas the black points and lines indicate individuals of Fabaceae in each site. The dotted line represents the MA-fitted line for all 1909 samples together. Asterisks indicate significance as in Fig. 1.

Sarmiento et al., 2011 (AJB)



Dalling et al., 2020 (AREES)



Zalamea et al., 2023 NSF grant (awarded)

Exploratory Data Analysis

“In Statistics [$\sim$ Data Science], EDA is an **approach** to analyzing data sets to summarize their main **characteristics**, often with visual methods.”



- Discover patterns
- Extract important variables
- Detect outliers and anomalies
- Check assumptions



Start working now!

Data exploration starts before going to the field/lab.

Think about your questions:

- * Possible answers?
- * What are you going to measure?
- * How?
- * Statistical analysis?

Record your data (basics)



Excel spreadsheets are an easy way to enter and record your data



Very popular (lots of users)

Easy to access

Some useful/easy tools



Easy to mess up your data

Not a statistical software

Not very flexible (advanced analyses)

Record your data (basics)



- * Be consistent and organized
(with files, naming variables, etc.)

Record your data (basics)



- * Be consistent and organized
(with files, naming variables, etc.)
- * Keep your spreadsheets simple
(think saving as .CSV)

Be careful!

Record your data (basics)



- * Be consistent and organized
(with files, naming variables, etc.)
- * Keep your spreadsheets simple
- * Record one "thing" per column

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- * Record one "thing" per column
- * Do not leave empty cells (0?, NA?)

Record your data (basics)



- * Be consistent and organized (with files, naming variables, etc.)
- * No special characters (or spaces)
- * Keep your spreadsheets simple
- * Record one "thing" per column
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- * No special characters (or spaces)
- * Name variables in an easy, clear way (Think R)

Record your data (basics)



- * Be consistent and organized (with files, naming variables, etc.)
- * Keep your spreadsheets simple
- * Record one "thing" per column
- * Do not leave empty cells (0?, NA?)
- * No special characters (or spaces)
- * Name variables in an easy, clear way (Think R)
- * Record metadata (another spreadsheet)

	A	B	C	D	E	F	G	H	I	J	K	L
1	TANK	1	2	3	4	5	6	7	8	9	10	11
2	[ATZ]	50	500	5	50	5	0	50	0	0	500	5
3	rep	2	4	6	6	2	6	1	5	1	6	4
4	Date	APRIL 24, 2002										
5	MASS	0.1568	0.0827	0.0641	0.0739	0.1433	0.0695	0.1708	0.1456	0.1727	0.1427	0.0723
6	DATE	8-May-02										
7	mass	0.3938	0.2044	0.1873	0.2547	0.4641	0.1871	0.3750	0.3481	0.3660	0.3758	0.2256
8	date	22-May-02										
9	Mass/larva	0.7551	0.4876	0.4102	0.4898	0.9739	0.3160	0.7240	0.6678	0.6855	0.6832	0.4418
10												
11	Density		24									
12			48									

Which one is better?

A	B	C	D	E	F
Tank #	Atrazine Conc. (ppb)	Density (48=48, 24=24)	Rep./Block	Date	Mass/larva
1	50	24	2	24-Apr-02	0.1568
2	500	48	4	24-Apr-02	0.0827
3	5	48	6	24-Apr-02	0.0641
4	50	48	6	24-Apr-02	0.0739
5	5	48	2	24-Apr-02	0.1433
6	0	48	6	24-Apr-02	0.0695
7	50	24	1	24-Apr-02	0.1708
8	0	24	5	24-Apr-02	0.1456
9	0	24	1	24-Apr-02	0.1727
10	500	24	6	24-Apr-02	0.1427
11	5	48	4	24-Apr-02	0.0723
12	50	48	5	24-Apr-02	0.0800
13	5	24	6	24-Apr-02	0.1308
14	0	48	4	24-Apr-02	0.0697
15	500	24	3	24-Apr-02	0.1511
16	500	24	1	24-Apr-02	0.1384

	A	B	C	D	E	F	G	H	I	J	K	L
1	TANK	1	2	3	4	5	6	7	8	9	10	11
2	[ATZ]	50	500	5	50	5	0	50	0	0	500	5
3	rep	2	4	6	6	2	6	1	5	1	6	4
4	Date	APRIL 24, 2002										
5	MASS	0.1568	0.0827	0.0641	0.0739	0.1433	0.0695	0.1708	0.1456	0.1727	0.1427	0.0723
6	DATE	8-May-02										
7	mass	0.3938	0.2044	0.1873	0.2547	0.4641	0.1871	0.3750	0.3481	0.3660	0.3758	0.2256
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9	Mass/larva	0.7551	0.4876	0.4102	0.4898	0.9739	0.3160	0.7240	0.6678	0.6855	0.6832	0.4418
10												
11	Density		24									
12			48									

- ★ One “thing” per column
- ★ Do not use cell color as data
- ★ Consistent format
- ★ Record your metadata
(e.g. mass/larva? units?)

A	B	C	D	E	F
Tank #	Atrazine Conc. (ppb)	Density (48=48, 24=24)	Rep./Block	Date	Mass/larva
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	A	B	C	D	E	F	G	H	I	J	K	L
1	TANK	1	2	3	4	5	6	7	8	9	10	11
2	[ATZ]	50	500	5	50	5	0	50	0	0	500	5
3	rep	2	4	6	6	2	6	1	5	1	6	4
4	Date	APRIL 24, 2002										
5	MASS	0.1568	0.0827	0.0641	0.0739	0.1433	0.0695	0.1708	0.1456	0.1727	0.1427	0.0723

Metadata:

“a set of data that describes and gives information about other data”

Any information that is useful for understanding your variables
(for others and for you in the future!)

- ★ One “thing” per column
- ★ Do not use cell color as data
- ★ Consistent format
- ★ Record your metadata
(e.g. mass/larva? units?)

A	B	C	D	E	F
Tank #	Atrazine Conc. (ppb)	Density (48=48, 24=24)	Rep./Block	Date	Mass/larva
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15	500	24	3	24-Apr-02	0.1511
16	500	24	1	24-Apr-02	0.1384

Record your data (basics)



Once your dataset is complete:

Save it and work on a copy

.csv only stores the current datasheet
(no colors, no filters, no formulas)

A real-life example:

The “old seeds” project (REU):
A sad story with a happy ending!

See excel file

A real-life example:

AutoSave

Bad_Example_Old_seeds

Home

Insert

New Tab

Draw

Page Layout

Formulas

Data

Review

View

Automate

Acrobat

Calibri (Body)

12

A1

Species

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
1	Species	Time	Germination	Germ_TZ_dormant	Germ_TZ_non_viable	Total_germ	TZ_viable	Z_non_viable	TZ_tot	SFR	Conductivity	Initial_weight	Final_weight	Permeability				
2	Trema black 2	0	30	151	16	197	183	16	199	9.76	0	2.84	3.00	0.05				
3	Trema black 2	1	75	77	24	176	113	71	184	9.54	0	3.20	3.23	0.01		408	287	60
4	Trema black 2	3	139	41	15	195	165	35	200	8.79	0	2.97	3.34	0.11		755	608	164
5	Trema black 2	6	164	18	5	187	147	42	189		0	3.05	3.09	0.02		0.5403974	0.4720395	0.3658537
6	Trema black 2	12	187	3	7	197	168	32	200		0	2.95	3.26	0.10				
7	Trema black 2	OLD	116	2		152												
8	Trema black 2	OLDER	43	1		58												
9	Zanthoxylum ekmanii	0	0	55	105	160	93	107	200	74.89	0	15.30	17.35	0.12				
10	Zanthoxylum ekmanii	1	26	52	115	193	97	96	193	68.32	0	16.24	17.66	0.08		145	140	454
11	Zanthoxylum ekmanii	3	54	27	108	189	62	137	199	60.02	0	15.88	17.15	0.07		739	314	478
12	Zanthoxylum ekmanii	6	65	6	126	197	62	138	200	61.50	0	15.61	16.76	0.07		0.1962111	0.4458599	0.9497908
13	Zanthoxylum ekmanii	12	76	4	112	192	83	116	199		0	14.59	15.75	0.07				
14	Zanthoxylum ekmanii	18	65	5	126	196	65	135	200		0	14.16	15.55	0.09				
15	Zanthoxylum ekmanii	OLD	38	4		102												
16	Zanthoxylum ekmanii	Oldest	2	0		43												
17																		
18																		
19		Time	% Germ	% Viable			Species	Age	Germination	Dormant	Inviabile		Species	Age	Germination	Dormant	Inviabile	
20		0	0.15228426	0.766497462			TB	0	0.152284264	0.766497	0.081218274		TB	<1	0.5403974	0.4720395	0.3658537	
21		1	0.42613636	0.4375			TB	1	0.426136364	0.4375	0.136363636		TB	1	0.9492386	0.015228426	0.035532995	
22		3	0.71282051	0.21025641			TB	3	0.712820513	0.210256	0.076923077		ZE	<1	0.1962111	0.4458599	0.9497908	
23		6	0.87700535	0.096256684			TB	6	0.877005348	0.096257	0.026737968		ZE	1	0.3958333	0.020833333	0.583333333	
24		12	0.94923858	0.015228426			TB	12	0.949238579	0.015228	0.035532995		ZE	1.5	0.3316327	0.0255102	0.6428571	
25		OLD	0.76315789	0.013157895			Species	Age	Germination	Dormant	Inviabile							
26		Older	0.74137931	0.017241379			ZE	0	0	0.34375	0.65625							
27							ZE	1	0.134715026	0.26943	0.595854922							
28							ZE	3	0.285714286	0.142857	0.571428571							
29							ZE	6	0.329949239	0.030457	0.639593909							
30							ZE	12	0.395833333	0.020833	0.583333333							
31							ZE	18	0.331632653	0.02551	0.642857143							
32														739				
33																		
34																		

A real-life example:

The happy ending!
2015 - 2023

Functional Ecology



RESEARCH ARTICLE | Free Access

Decadal survival of tropical pioneer seeds in the soil seed bank is accompanied by fungal infection and dormancy release

Paul-Camilo Zalamea , Carolina Sarmiento, A. Elizabeth Arnold, Venus Kuo, Carolyn Delevich, Adam S. Davis, Thomas A. Brown, James W. Dalling

First published: 14 December 2023 | <https://doi.org/10.1111/1365-2435.14476>

Take an initial look at your dataset



1 Dimensions

- ☐ Number of columns (**variables**) looks ok?
- ☐ Number of rows (**individuals/experimental units**) looks ok?

Take an initial look at your dataset



1 Dimensions

- ☐ Number of columns (**variables**) looks ok?
- ☐ Number of rows (**individuals/experimental units**) looks ok?

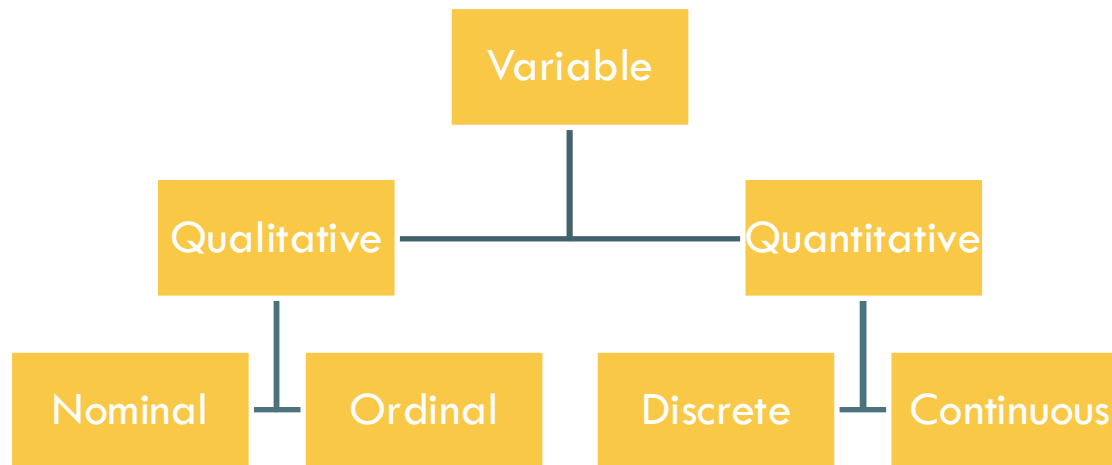
*** Depends on your experiment ***

Take an initial look at your dataset



2 Variables

Categorical or continuous?



Take an initial look at your dataset



2 Variables

Qualitative

~ not countable, categorical

Ordinal: If there is an order (e.g. Low - Medium – High)

Nominal: no particular order (e.g. Female - Male)

Take an initial look at your dataset



2 Variables

Quantitative

~ Things you can measure (numerical)

Discrete: Things you can count (e.g. species in a plot)

Continuous: ~ infinite (e.g. body mass)

Take an initial look at your dataset



2 Variables

Categorical or continuous?

How many in your dataset?

If categorical, do your categories look OK?

Take an initial look at your dataset

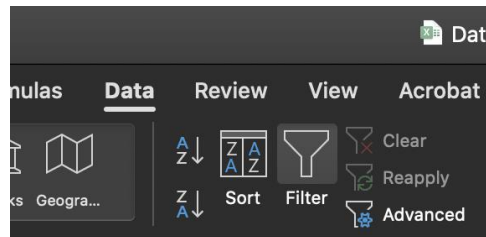


2 Variables

Categorical or continuous?

How many in your dataset?

If categorical, do your categories look OK?



Tip: Use the filter tool

(Mac: command + shift + F)

(Windows: control + shift + L) ?

Take an initial look at your dataset



	A	B	C	D	E	F	G	H	I	J	K	L	M	N
1	fun_code	seed_id	sequence	plate	slant_set	sp_init	species	time	plot	rep	species			
2	PS0001	AM0_25	CGGGCAGCT	343	1	AM	Apeiba_men	0	NA	NA				
3	PS0002	AM0_67	AGTTAATCCA	343	1	AM	Apeiba_men	0	NA	NA				
4	PS0003	AM0_74	GCTTTGGCG	343	1	AM	Apeiba_men	0	NA	NA				
5	PS0004	AM0_76	TACCGAGTT	343	1	AM	Apeiba_men	0	NA	NA				
6	PS0005	AM0_77	CTTTGGCGG	398	1	AM	Apeiba_men	0	NA	NA				
7	PS0006	AM0_85	TCGGGCGCAG	343	1	AM	Apeiba_men	0	NA	NA				
8	PS0007	AM0_90	GCTTTGGCG	343	1	AM	Apeiba_men	0	NA	NA				
9	PS0008	AM0_93	CTTTGGCGG	343	1	AM	Apeiba_men	0	NA	NA				
10	PS0009	AM0_99	TTTGGCGGC	343	1	AM	Apeiba_men	0	NA	NA				
11	PS0010	AM0_103	ATCGATGAA	343	1	AM	Apeiba_men	0	NA	NA				
12	PS0011	AM0_104	TACCGAGTT	343	1	AM	Apeiba_men	0	NA	NA				
13	PS0012	AM0_108	TTTGGCGGC	343	1	AM	Apeiba_men	0	NA	NA				
14	PS0013	AM0_109	TTATAGGTC	343	1	AM	Apeiba_men	0	NA	NA				
15	PS0014	AM0_110	TACCGAGTT	343	1	AM	Apeiba_men	0	NA	NA				
16	PS0017	AM0_123	TTTGGCGGC	343	1	AM	Apeiba_men	0	NA	NA				
17	PS0018	AM0_125	CCCGCTCCCG	343	1	AM	Apeiba_men	0	NA	NA				
18	PS0019	AM0_126	CTTTGGCGG	343	1	AM	Apeiba_men	0	NA	NA				
19	PS0020	AM0_138	TTATCGTGGA	350	1	AM	Apeiba_men	0	NA	NA				
20	PS0022	AM1A1_2	TACCGAGTT	343	1	AM	Apeiba_men	1	A					
21	PS0024	AM1A3_1	AACTCCAACCG	350	1	AM	Apeiba_men	1	A					
22	PS0025	AM1A3_5	GGATCATTAA	343	1	AM	Apeiba_men	1	A					
23	PS0028	AM1D2_2	TTCGGGCTT	343	1	AM	Apeiba_men	1	D					
24	PS0031	AM1H1_2	GAGTTATCA	343	1	AM	Apeiba_men	1	H					
25	PS0035	AM1H3_1	TATTACCCTT	343	1	AM	Apeiba_men	1	H					
26	PS0036	AM1H3_10	GTTATAGGT	343	1	AM	Apeiba_men	1	H					
27	PS0037	AM1H3_4	TTACCGACT	343	1	AM	Apeiba_men	1	H					

species

Sort

A

Ascending

Z

Descending

By color:

None

Filter

By color:

None

Choose One

Q

Search

☒

(Select All)

☒

Annona_spraguei

☒

Apeiba_membranace

☒

Apeiba_membranacea

☒

Apeiba_tibourbou

☒

Cecropia_insignis

☒

Cecropia_longipes

☒

Cecropia_peltata

Clear Filter

NA

0

23

18

NA

0

24

10

Take an initial look at your dataset



	A	B	C	D	E	F	G	H	I	J	K	L	M	N	
1	fun_code	seed_id	sequence	plate	slant_set	sp_init	species	time	plot	rep	species				
2	PS0001	AM0_25	CGGGCAGCT	343	1	AM	Apeiba_men	0	NA	NA					
3	PS0002	AM0_67	AGTTAATCCA	343	1	AM	Apeiba_men	0	NA	NA					
4	PS0003	AM0_74	GCTTTGGCGG	343	1	AM	Apeiba_men	0	NA	NA					
5	PS0004	AM0_76	TACCGAGTT	343	1	AM	Apeiba_men	0	NA	NA					
6	PS0005	AM0_77	CTTTGGCGG	398	1	AM	Apeiba_men	0	NA	NA					
7	PS0006	AM0_85	TCGGGCGCAG	343	1	AM	Apeiba_men	0	NA	NA					
8	PS0007	AM0_90	GCTTTGGCGG	343	1	AM	Apeiba_men	0	NA	NA					
9	PS0008	AM0_93	CTTTGGCGG	343	1	AM	Apeiba_men	0	NA	NA					
10	PS0009	AM0_99	TTTGGCGGGC	343	1	AM	Apeiba_men	0	NA	NA					
11	PS0010	AM0_103	ATCGATGAA	343	1	AM	Apeiba_men	0	NA	NA					
12	PS0011	AM0_104	TACCGAGTT	343	1	AM	Apeiba_men	0	NA	NA					
13	PS0012	AM0_108	TTTGGCGGGC	343	1	AM	Apeiba_men	0	NA	NA					
14	PS0013	AM0_109	TTATAGGTC	343	1	AM	Apeiba_men	0	NA	NA					
15	PS0014	AM0_110	TACCGAGTT	343	1	AM	Apeiba_men	0	NA	NA					
16	PS0017	AM0_123	TTTGGCGGGC	343	1	AM	Apeiba_men	0	NA	NA					
17	PS0018	AM0_125	CCCGCTCCCG	343	1	AM	Apeiba_men	0	NA	NA					
18	PS0019	AM0_126	CTTTGGCGG	343	1	AM	Apeiba_men	0	NA	NA					
19	PS0020	AM0_138	TTATCGTGG	350	1	AM	Apeiba_men	0	NA	NA					
20	PS0022	AM1A1_2	TACCGAGTT	343	1	AM	Apeiba_men	1	A		1				
21	PS0024	AM1A3_1	AACTCCAACC	350	1	AM	Apeiba_men	1	A		3				
22	PS0025	AM1A3_5	GGATCATT	343	1	AM	Apeiba_men	1	A		3				
23	PS0028	AM1D2_2	TTCGGGCTT	343	1	AM	Apeiba_men	1	D		2				
24	PS0031	AM1H1_2	GAGTTATCA	343	1	AM	Apeiba_men	1	H		1				
25	PS0035	AM1H3_1	TATTACCCTT	343	1	AM	Apeiba_men	1	H		3				
26	PS0036	AM1H3_10	GTTATAGGT	343	1	AM	Apeiba_men	1	H		3	NA	0	23	18
27	PS0037	AM1H3_4	TTACCGACT	343	1	AM	Apeiba_men	1	H		3	NA	0	24	18

species

Sort

A ↓ Ascending Z ↓ Descending

By color: None

Filter

By color: None

Choose One

Q Search

☒ (Select All)

☒ Annona_spraguei

☒ Apeiba_membranace

☒ Apeiba_membranacea

☒ Apeiba_tibourbou

☒ Cecropia_insignis

☒ Cecropia_longipes

☒ Cecropia_peltata

Clear Filter

This typo will introduce two different categories for the same species!

Take an initial look at your dataset



3 **Values**

Do not leave empty cells

Do not leave empty cells



Fill them with
"NA" if it is a
true missing
value

Take an initial look at your dataset



3 **Values**

Do not leave empty cells

Do your values make sense?

Take an initial look at your dataset



3 Values

Do not leave empty cells

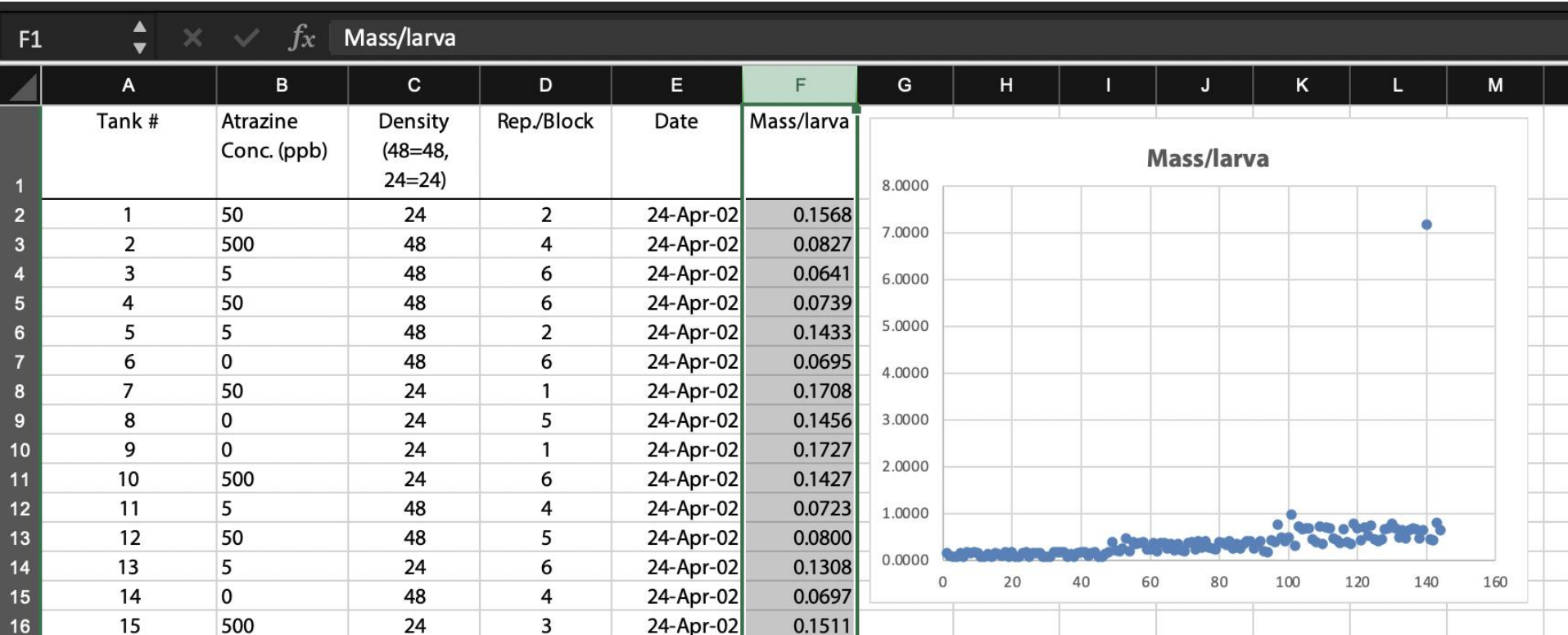
Do your values make sense?



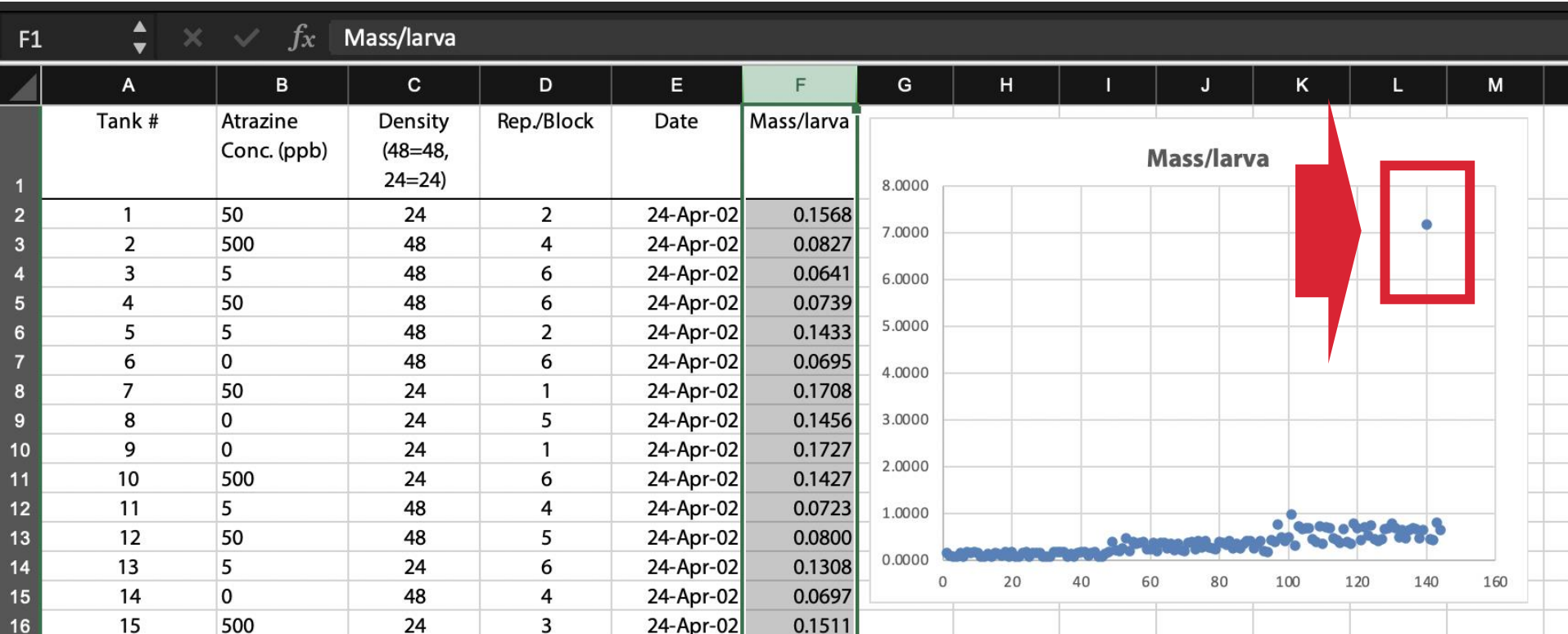
Tricky question!

Typing mistakes? Outliers? Duplicated values?

Simple visualizations are useful!

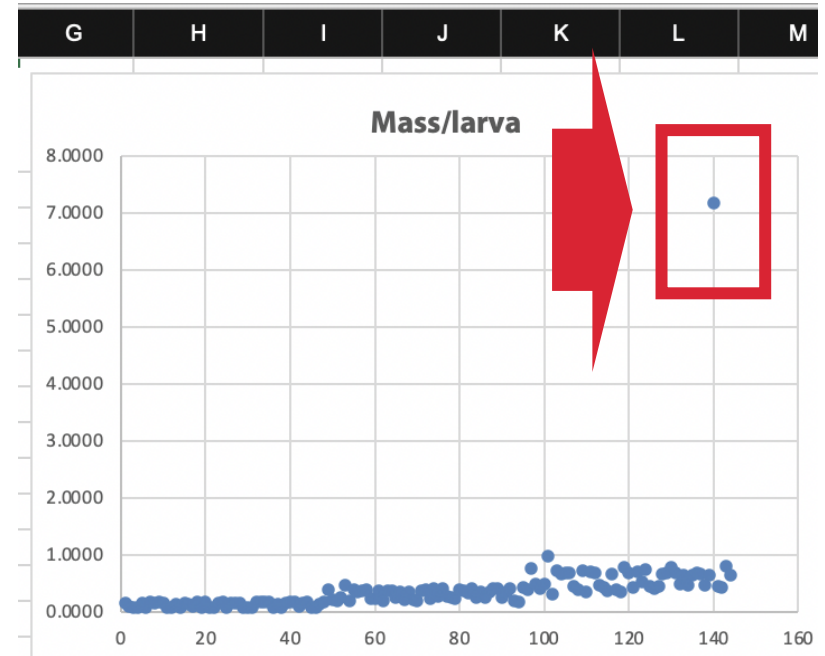


Simple visualizations are useful!



Simple visualizations are useful!

- ☐ Go back to your lab/field notes
- ☐ Correct – only if you are 100% sure it is a typo
- ☐ Do not change it if you are not sure
- ☐ Is it a measurement error?
Make a note, but do not erase it



Take an initial look at your dataset



3 Values

Do not leave empty cells

Do your values make sense?



Tricky question!

Duplicated values?

Tip: Use conditional formatting

Take an initial look at your dataset



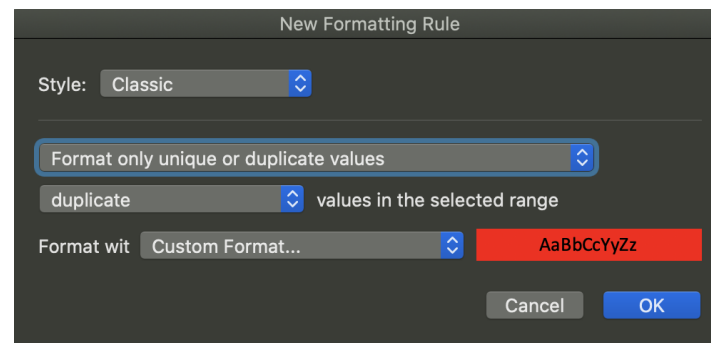
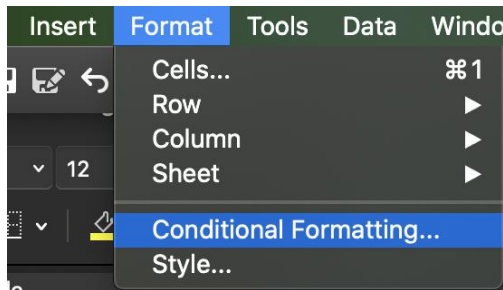
3 Values

Do not leave empty cells

Do your values make sense?



Tricky question!



PS0037	AM1H3_4	TTACCGAGT	343
PS0038	AM1H4_10	TTATCACAAC	394
PS0041	AM1P3_1	ACAACGGAT	402
PS0041	AM1P3_1	GCGGAGGG	625
PS0042	AM1P3_2	TGCTTTGGC	343

Take an initial look at your dataset



3 Values

Do some basic calculations, to help you understand your data:

- Max. value [= max()]
- Min. value [= min()]
- Mean [= average()]
- Standard deviation [= stdev()]

Take an initial look at your dataset



4 Distribution of data:

- Is a *function* that shows all the possible values of a variable - it also tells you how often each value occurs.

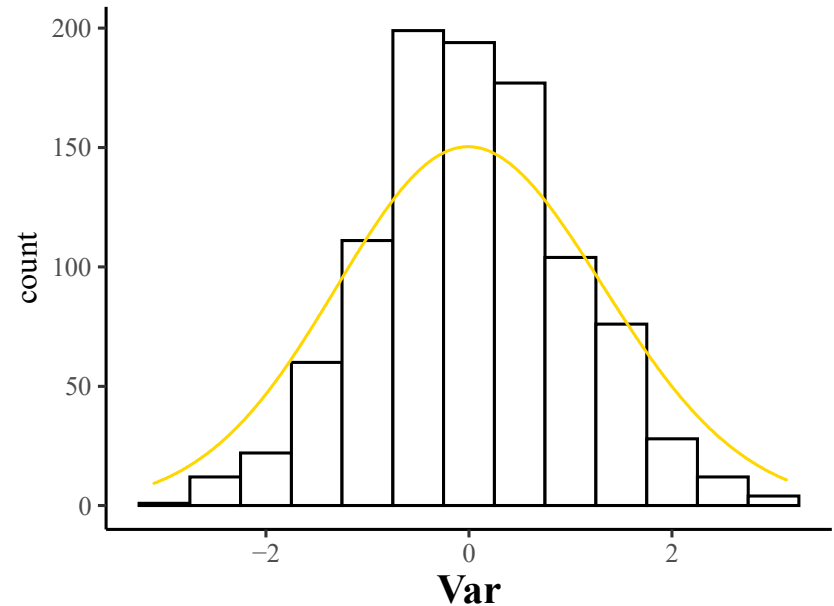
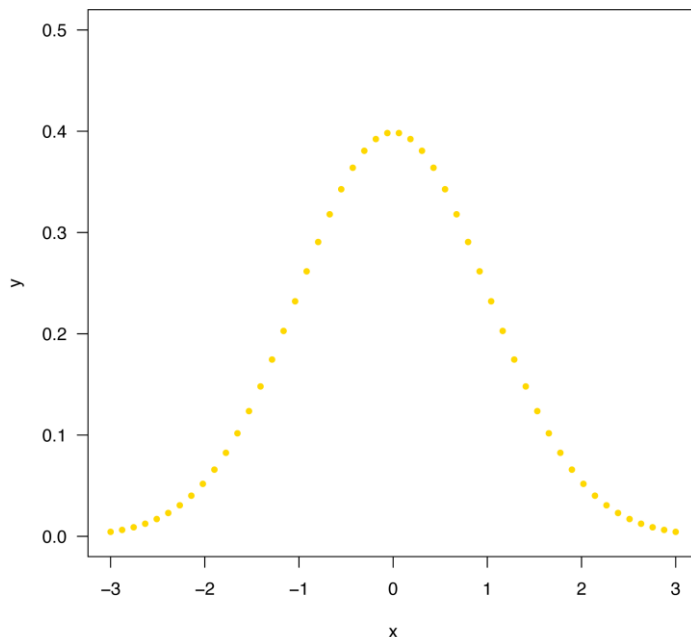
The shape of the graph when all possible values are plotted on a frequency graph or frequency distribution histogram

- Hypothesis tests are based on specific distributional assumptions (often, does the variable follow a normal distribution?)
- Needs to be verified so that the statistical technique yields valid conclusions.

Take an initial look at your dataset



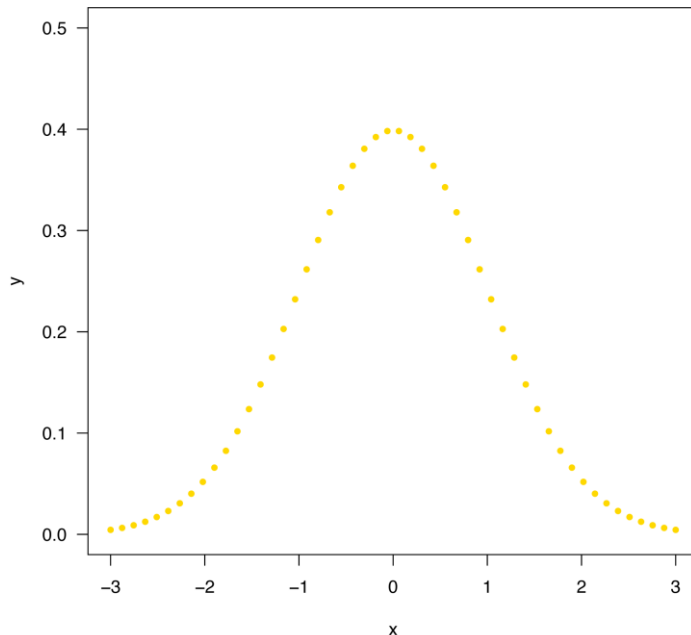
Normal Distribution (Gaussian):



Take an initial look at your dataset



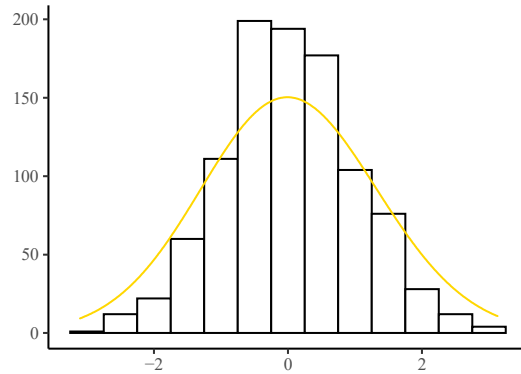
Normal Distribution (Gaussian):



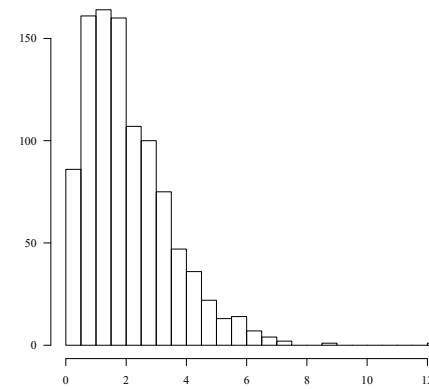
- ☐ Mean, mode and median coincide with each other
- ☐ The distribution has a bell-shaped distribution curve
- ☐ The distribution curve is symmetrical to the center
- ☐ The area under the curve is equal to 1

Continuous distributions:

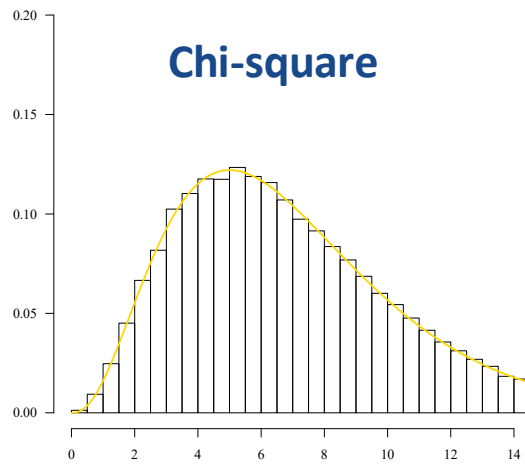
Normal



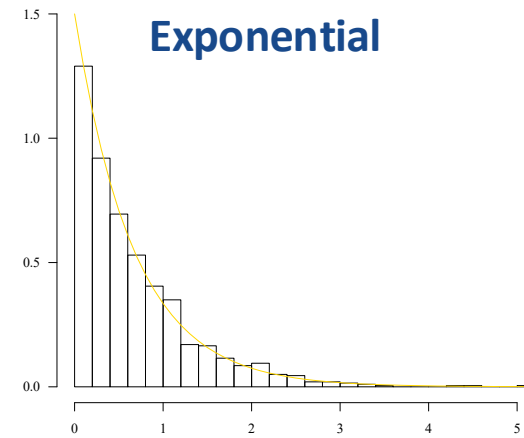
Gamma



Chi-square



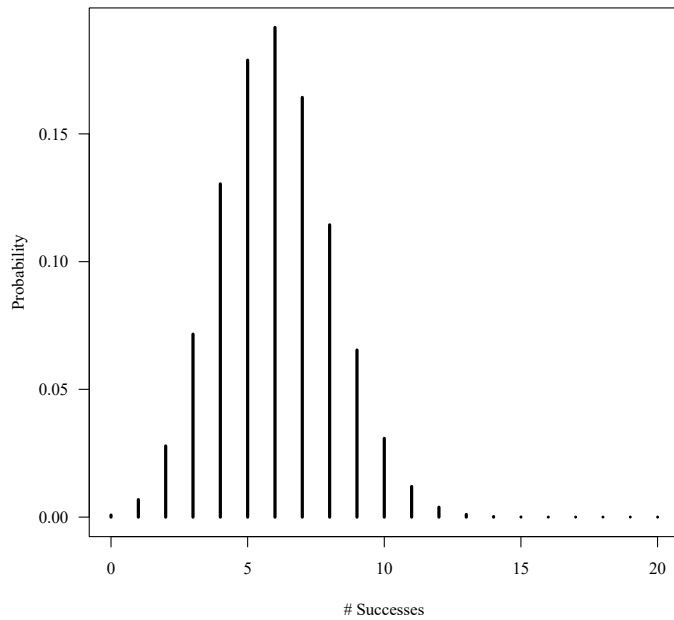
Exponential



Discrete distributions:

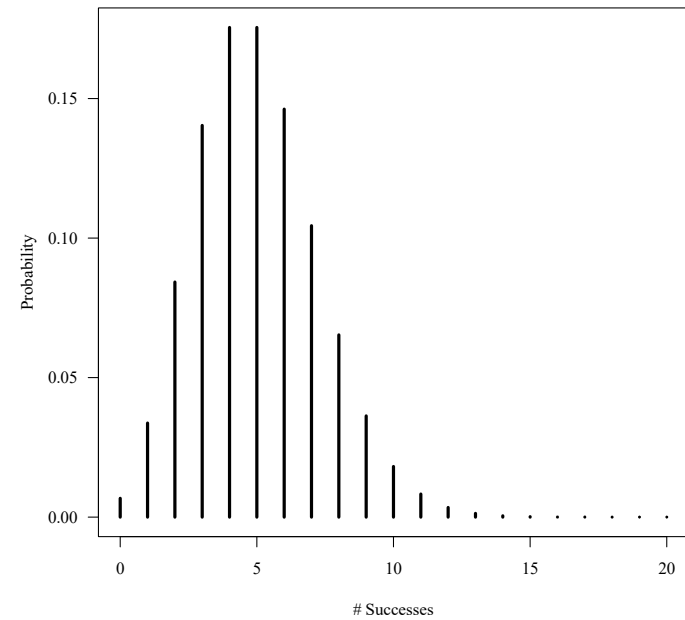
Binomial

Binomial Distribution ($n=20$, $p=0.3$)



Poisson

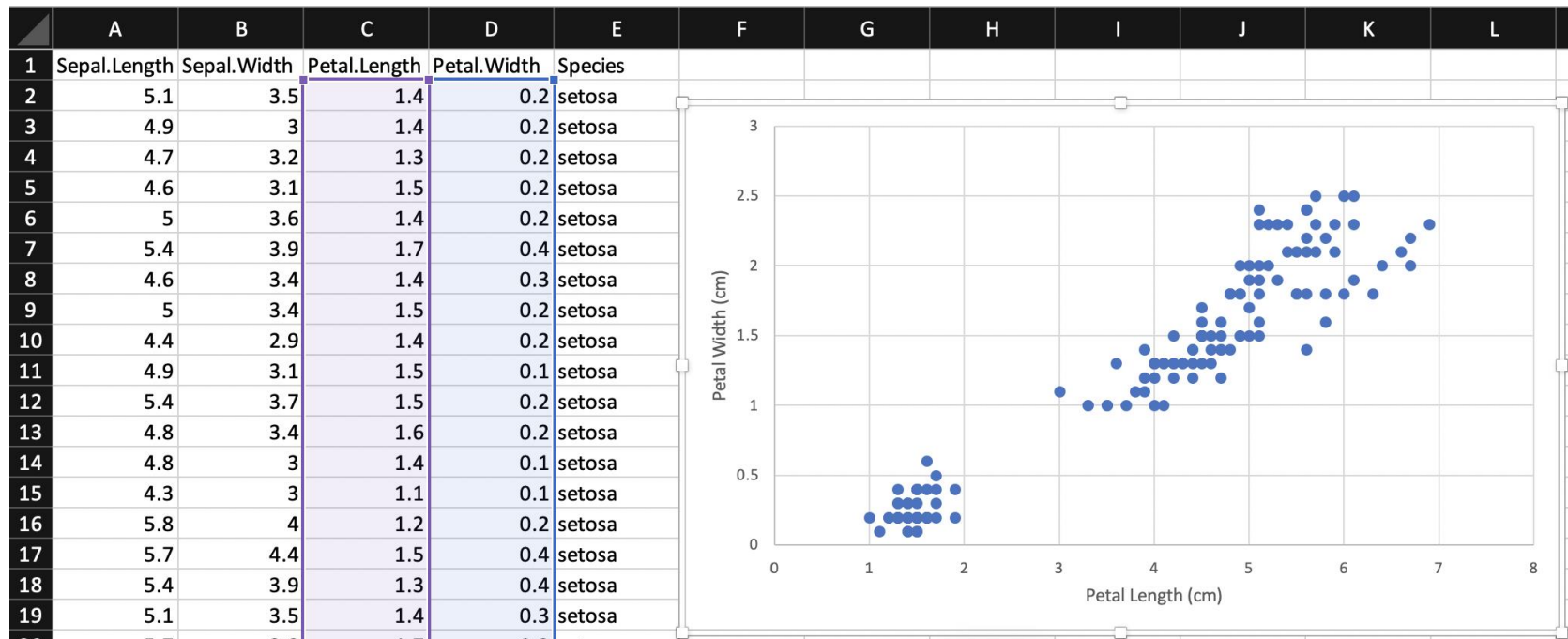
Poisson Distribution ($\lambda = 5$)



Start looking for patterns



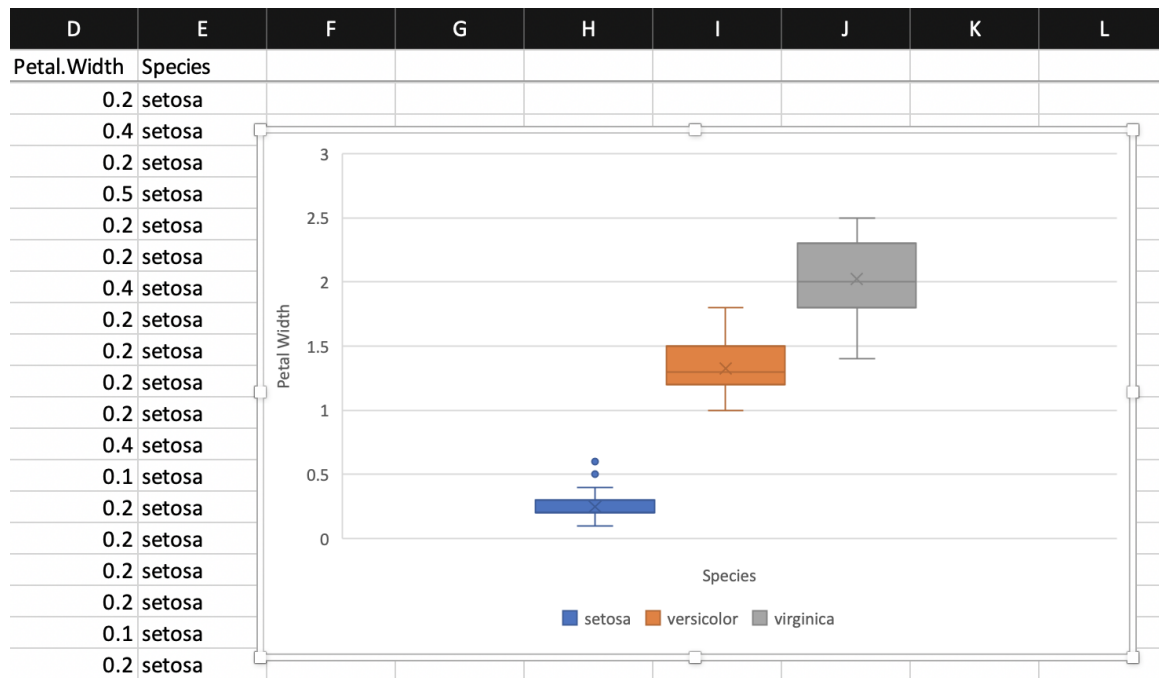
Are some of your variables correlated?



Start looking for patterns



Are there (visual) differences among
your treatments/species?



Start looking for patterns

(practice at home 🧐)

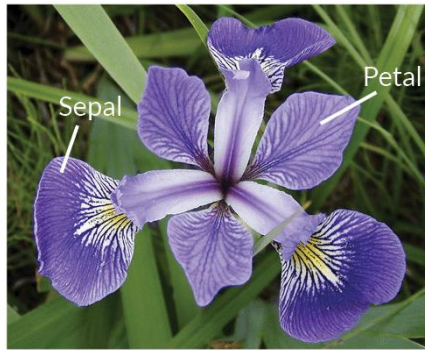


1. Are there changes in larval mass through time?
2. Does the atrazine concentration affect the size of the larvae?
3. Does the larval density in the tanks affect larval size?
4. Which variable looks more normal - larval mass or log-transformed larval mass? [Hint](#)

Use only plots (Excel charts)

Start looking for patterns

(practice at home 🧐)



Iris versicolor



Iris setosa



Iris virginica

1. Which species has the largest flowers?
2. Which floral structure is smaller, petals or sepals?
3. Which variable looks more normal?

Use only Excel plots (charts)

Introduction to

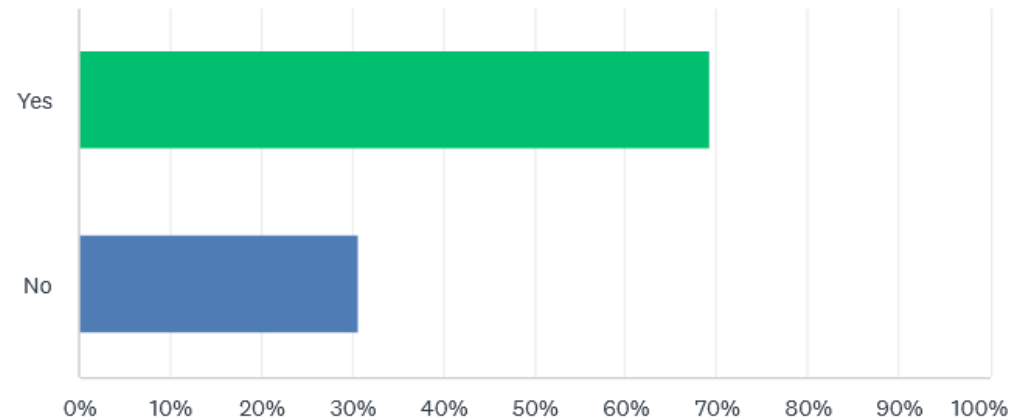
Carolina Sarmiento
USF
October 9, 2026



The survey:

Have you used R before?

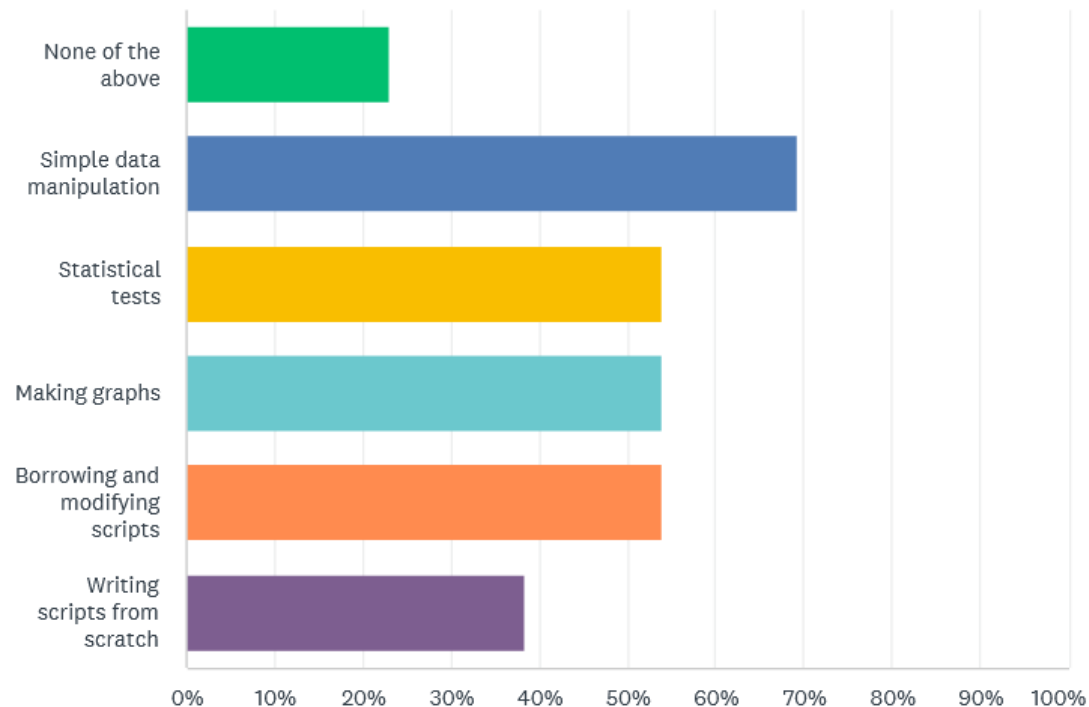
Answered: 13 Skipped: 0



The survey:

Which things do you feel comfortable doing with R? Select all that apply

Answered: 13 Skipped: 0



What is R?

“R is a **language** and **environment** for statistical computing and graphics.”

- “To use it well, one must master its grammar and vocabulary.”
- It allows users to define new functions (= add additional functionality) – as opposed to very specific and inflexible tools offered by other data analysis software.
- “Provides a wide variety of statistical and graphical techniques and is highly extensible.”

It's a **free** and **open-source** software

Open-source?

Software for which the original source code is made freely available and may be redistributed and modified

- Provides full access to algorithms and their implementation
- Allows researchers to explore and expand the methods used to analyze data
- Promotes reproducible research by providing open and accessible tools
- Most of R is written in... R! This makes it easier to see what functions are actually doing

Why use R?

- It's **free** (and it won't change)
- State of the art – cutting edge **packages** are well ahead of commercial software
[Publication example](#) or [list of packages by date](#)
- Great **graphic** capabilities
- Active user **community** – Good online resources
- Forces you to **think** about analyses
- R is **stable**. It will continue to exist, even if programmers/users stop adding to it. Therefore, unlikely to go away.
- You can get it to **do exactly what you need** it to do:
Probably someone else has already done most of the work for you
- Almost every analytical **tool** you can think of is available (or will be soon)

But some disadvantages...

- **Not user friendly:** Steep learning curve, minimal GUI (graphical user interface)
- Quality control – depends on **volunteer** programmers
- No commercial support: You must figure out how to use a function or correct methods to implement it (it can be **frustrating**)
- Help files are **not user-friendly**: Learning how to read them is very important

Base R and packages

What is a package?

A collection of functions, documentation, and data sets developed by the community. Packages increase the power of R by improving existing base R functionalities, or by adding new ones.

The R base package contains (many) **basic functions** for R to work:

- arithmetic
- input/output
- basic programming support
- etc.

Other packages uses R base to do many other things:

- Natural Language Processing (~analysis of speech and language) of Korean texts
- Extract weather data from the web
- Estimate actual evapotranspiration using land surface energy balance models

Packages in CRAN:

16,395 (2020) / 18,287 (2021) / 18,689 (2022) / 19,955 (2023) / 21,594 (2024) / 22,859 (2025)

Repositories

**The location where the packages are stored
(typically, online and accessible to everyone)**

The 3 most popular:

1. CRAN (Comprehensive R Archive Network)

- Network of servers maintained by the R community
- R foundation coordinates it
- For a package to be published here, it needs to pass several tests to ensure it follows CRAN policies

2. Bioconductor:

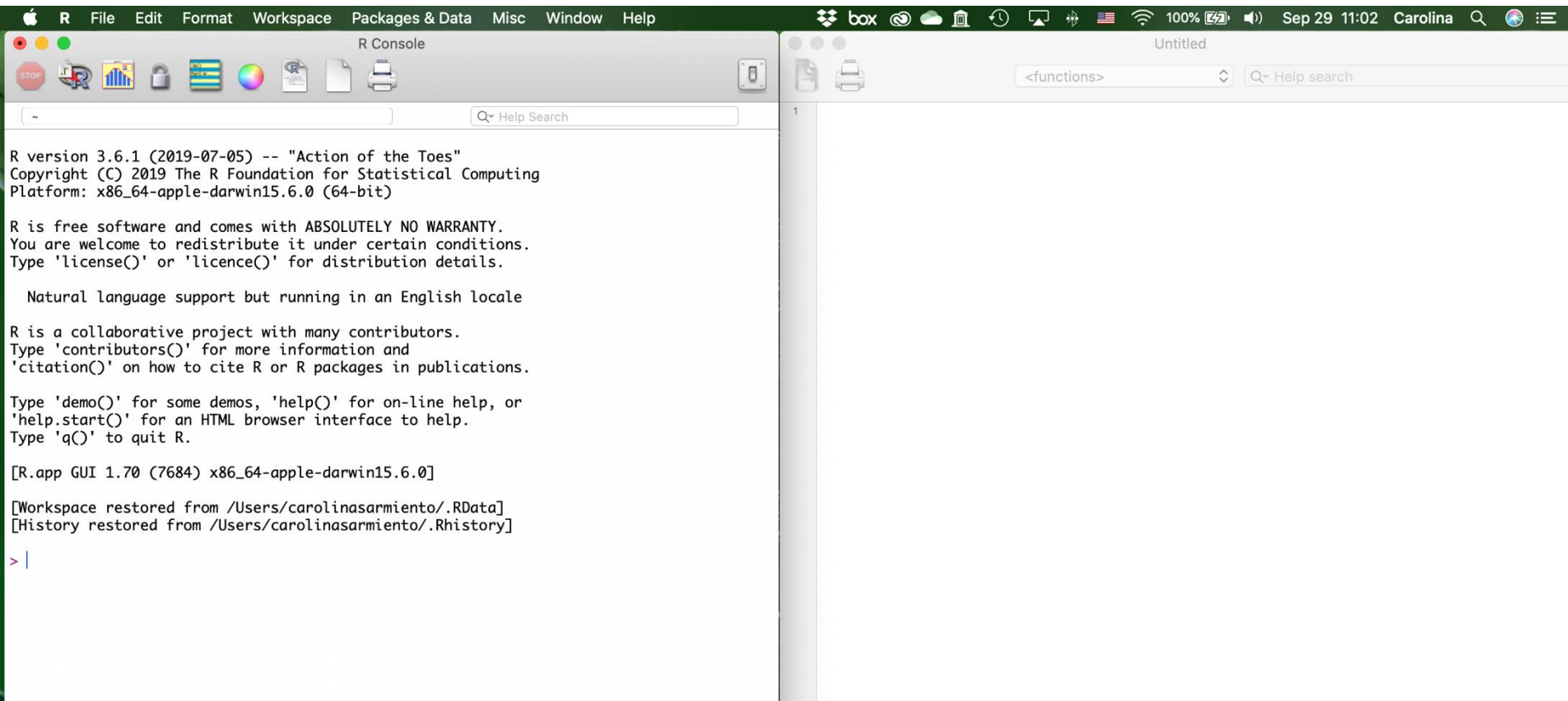
- Topic-specific repository, intended for open-source software for bioinformatics.
- It has its own submission and review processes
- Very active community (several conferences and meetings per year)

3. GitHub:

- Not R specific
- Probably the most popular repository for open-source projects
- Easy to share and collaborate with others
- No review process associated with it

Let's get into it!

Take a look:

A screenshot of the R Studio application window. The top menu bar includes Apple logo, R, File, Edit, Format, Workspace, Packages & Data, Misc, Window, and Help. The top toolbar contains icons for R, File, Edit, Format, Workspace, Packages & Data, Misc, Window, and Help. The main window is split into two panes. The left pane is the 'R Console' and contains the following text:

```
R version 3.6.1 (2019-07-05) -- "Action of the Toes"
Copyright (C) 2019 The R Foundation for Statistical Computing
Platform: x86_64-apple-darwin15.6.0 (64-bit)

R is free software and comes with ABSOLUTELY NO WARRANTY.
You are welcome to redistribute it under certain conditions.
Type 'license()' or 'licence()' for distribution details.

Natural language support but running in an English locale

R is a collaborative project with many contributors.
Type 'contributors()' for more information and
'citation()' on how to cite R or R packages in publications.

Type 'demo()' for some demos, 'help()' for on-line help, or
'help.start()' for an HTML browser interface to help.
Type 'q()' to quit R.

[R.app GUI 1.70 (7684) x86_64-apple-darwin15.6.0]

[Workspace restored from /Users/carolinasarmiento/.RData]
[History restored from /Users/carolinasarmiento/.Rhistory]

> |
```

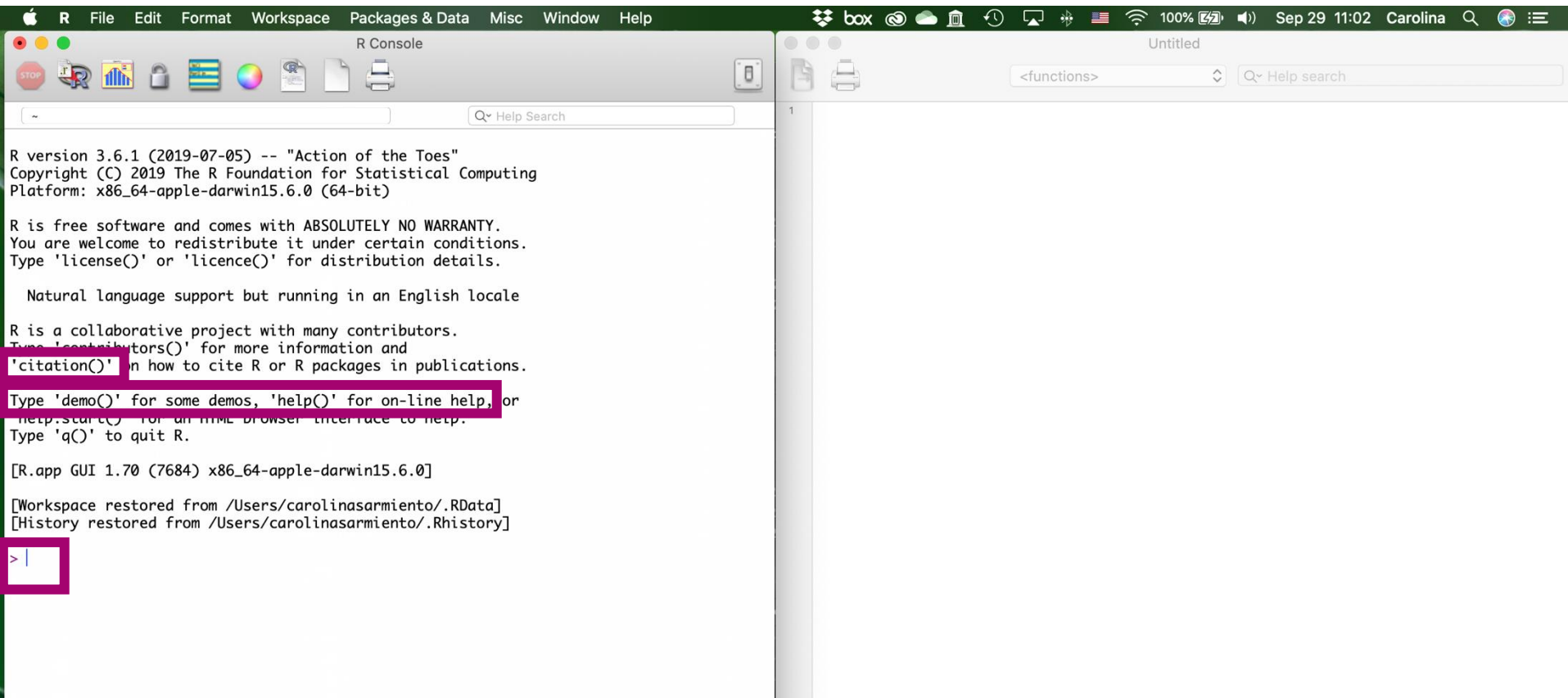
The right pane is titled 'Untitled' and contains a search bar with the text '<functions>' and a 'Help search' button. The status bar at the bottom shows the date and time 'Sep 29 11:02' and the name 'Carolina'.

Let's get into it!

Console



Text Editor



Let's get into it!

Console



```
Type 'demo()' for some demos, 'help()' for on-line help, or  
'help.start()' for an HTML browser interface to help.  
Type 'q()' to quit R.
```

```
[R.app GUI 1.70 (7735) x86_64-apple-darwin15.6.0]
```

```
> 2+2  
[1] 4  
> Carolina  
Error: object 'Carolina' not found  
> |
```

Text Editor



```
Untitled  
<functions>  
1 15+(10*10)  
2  
3 "My name is Carolina"  
4  
5 My name is Carolina
```



Run line by line (cursor)

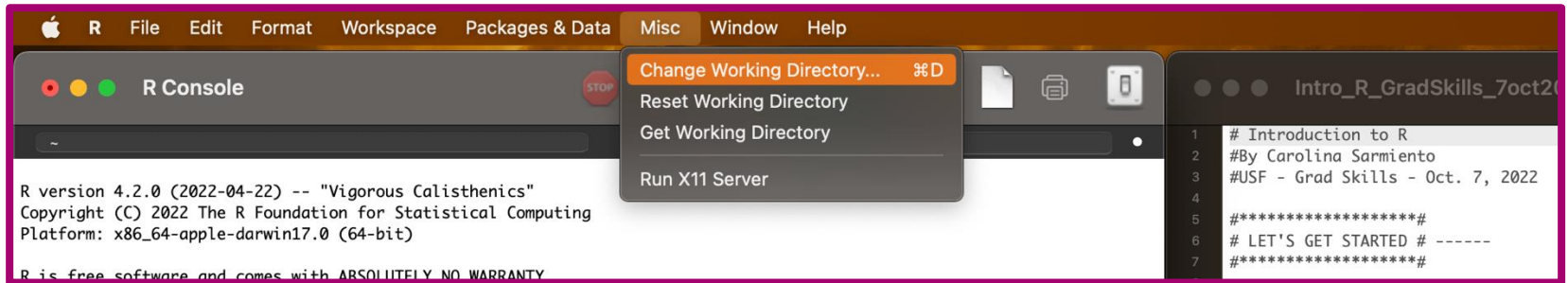
or

Highlight text to run

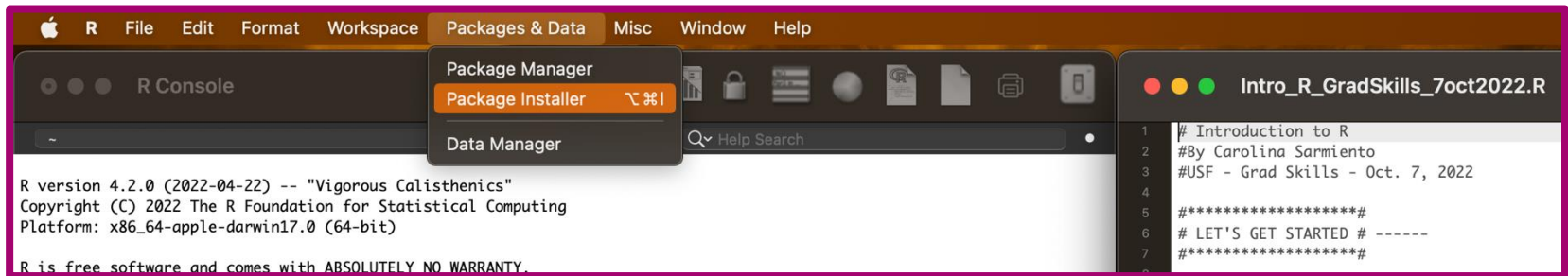
Mac: command + Enter

Win: ctrl + Enter

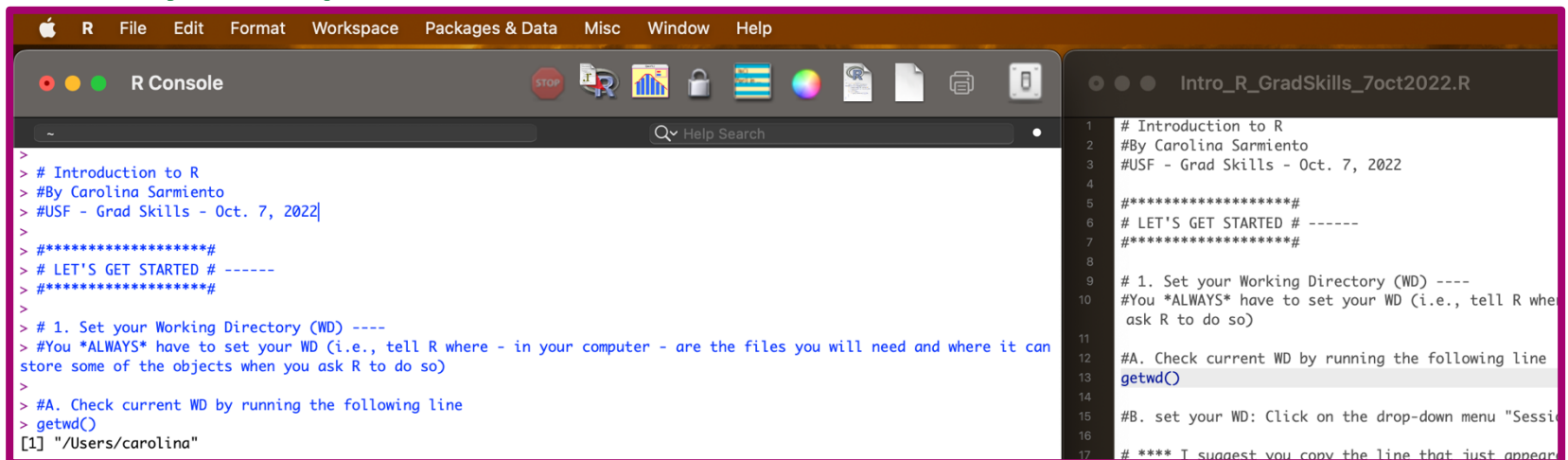
1. Set up the Working Directory



2. Install packages



3. Run your script





Open-source environment for working with R:

“RStudio integrates the tools you use with R into a single environment”

Includes graphical interface tools for many common R procedures

Includes:

- console
- syntax-highlighting editor
- tools for plotting
- R help
- history
- debugging
- workspace management



Quiescent_germ_fungal_8Aug2018...
dada_phyloseq_JCexp.R
Intro_R_GradSkills_7oct2022.R

1 # Introduction to R
2 #By Carolina Sarmiento
3 #USF - Grad Skills - Oct. 7, 2022
4
5 #*****#
6 # LET'S GET STARTED # -----
7 #*****#
8
9 # 1. Set your Working Directory (WD) -----
10 #You *ALWAYS* have to set your WD (i.e., tell R where - in your computer - are the files you will need and where it can store some of the
11 #objects when you ask R to do so)-
12
13 #A. Check current WD by running the following line-
14 getwd()
15
16 #B. set your WD: Click on the drop-down menu: "Session" -> Set Working Directory -> Choose Directory-
17
18 # **** I suggest you copy the line that just appeared in the Console to your script, for next time you open your script, just run that line-
19
20 setwd("~/Desktop/Grad_Skills/R") # copy yours here!
21
22 # 2. Install a package -----
23
24 # A. Install the package-
25 # Use first the "Install" button on the Packages tab, look for the package of interest (dplyr) and make sure to check "install dependencies"-
3:34 (Top Level) - R Script

Console Terminal Background Jobs
R 4.2.0 ~/ /
R version 4.2.0 (2022-04-22) -- "Vigorous Calisthenics"
Copyright (C) 2022 The R Foundation for Statistical Computing
Platform: x86_64-apple-darwin17.0 (64-bit)

R is free software and comes with ABSOLUTELY NO WARRANTY.
You are welcome to redistribute it under certain conditions.
Type 'license()' or 'licence()' for distribution details.

Natural language support but running in an English locale

R is a collaborative project with many contributors.
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'citation()' on how to cite R or R packages in publications.

Type 'demo()' for some demos, 'help()' for on-line help, or
'help.start()' for an HTML browser interface to help.
Type 'q()' to quit R.

Warning: macOS is reporting that you have not yet agreed to the Xcode license.
This can occur if Xcode has been updated or reinstalled (e.g. as part of a macOS update).
Some features (e.g. Git / SVN) may be disabled.

Please run:

sudo xcodebuild -license accept

in a terminal to accept the Xcode license, and then restart RStudio.

>

Environment History Connections
R - Global Environment
Environment is empty

Files Plots Packages Help Viewer Presentation
Home - Find in Topic

R Resources

Learning R Online
CRAN Task Views
R on StackOverflow
Getting Help with R

RStudio

RStudio IDE Support
RStudio Community Forum
RStudio Cheat Sheets
RStudio Tip of the Day
RStudio Packages
RStudio Products

Manuals

An Introduction to R
Writing R Extensions
R Data Import/Export

The R Language Definition
R Installation and Administration
R Internals

Reference

Packages

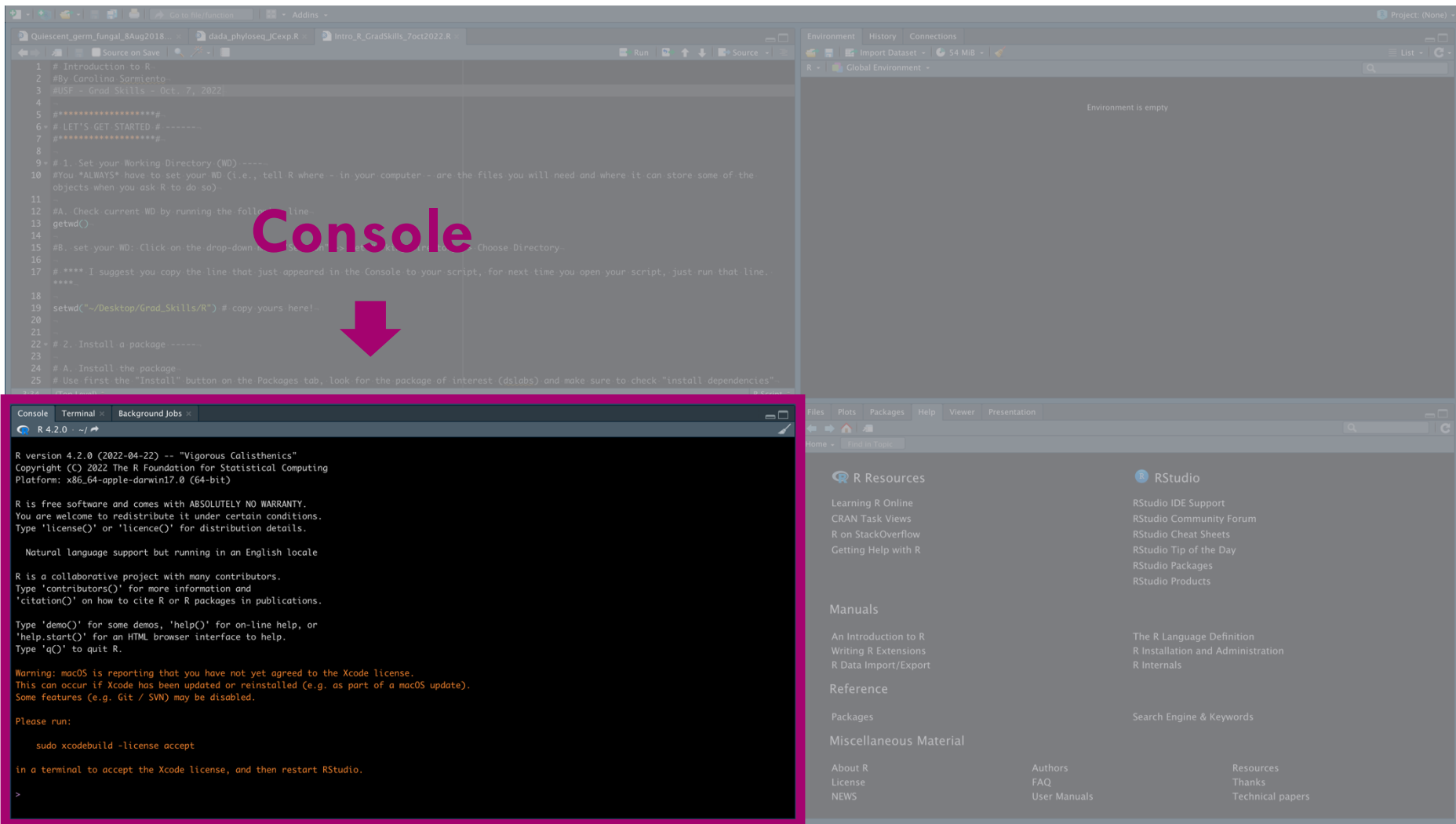
Search Engine & Keywords

Miscellaneous Material

About R
License
NEWS

Authors
FAQ
User Manuals

Resources
Thanks
Technical papers





The screenshot shows the R Studio interface. A pink box highlights the Source editor, which contains an R script. A pink arrow points from the Source editor to the Console, which displays the R version and license information.

Source editor (your script) and Data viewer

```
# Introduction to R
# By Carolina Sarmiento
# USF - Grad Skills - Oct. 7, 2022
# *****
# LET'S GET STARTED #
# *****
# 1. Set your Working Directory (WD) ----
# You *ALWAYS* have to set your WD (i.e., tell R where - in your computer - are the files you will need and where it can store some of the
# objects when you ask R to do so)
# A. Check current WD by running the following line:
getwd()
# B. set your WD: Click on the drop-down menu: "Session" -> Set Working Directory -> Choose Directory
# **** I suggest you copy the line that just appeared in the Console to your script, for next time you open your script, just run that line.
# ****
setwd("~/Desktop/Grad_Skills/R") # copy yours here!
# 2. Install a package -----
# A. Install the package
# Use first the "Install" button on the Packages tab, look for the package of interest (dslabs) and make sure to check "install dependencies"
```

R version 4.2.0 (2022-04-22) -- "Vigorous Calisthenics"
Copyright (C) 2022 The R Foundation for Statistical Computing
Platform: x86_64-apple-darwin17.0 (64-bit)

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Natural language support is running on an unsupported platform!
R is a collaborative project with many contributors.
Type 'contributors()' for more information and
'citation()' on how to cite R or R packages in publications.

Type 'demo()' for some demos, 'help()' for on-line help, or
'help.start()' for an HTML browser interface to help.
Type 'q()' to quit R.

Warning: macOS is reporting that you have not yet agreed to the Xcode license.
This can occur if Xcode has been updated or reinstalled (e.g. as part of a macOS update).
Some features (e.g. Git / SVN) may be disabled.

Please run:

```
sudo xcodebuild -license accept
```

in a terminal to accept the Xcode license, and then restart RStudio.

>

R Resources

- Learning R Online
- CRAN Task Views
- R on StackOverflow
- CRAN Task Views

RStudio

- RStudio IDE Support
- RStudio Community Forum
- RStudio Cheat Sheets
- RStudio News
- RStudio 1.0.159

Manuals

- An Introduction to R
- Writing R Extensions
- R Data Import/Export
- The R Language Definition
- R Installation and Administration
- R Internals

Reference

- Packages
- Search Engine & Keywords

Miscellaneous Material

- About R
- License
- NEWS
- Authors
- FAQ
- User Manuals
- Resources
- Thanks
- Technical papers



Environment



Shows the objects (i.e., data frames, arrays, values, and functions) you have in your workspace

The image shows a screenshot of the RStudio IDE. The main editor window on the left contains an R script with comments and code for setting the working directory and installing a package. The Environment pane on the right is titled 'Global Environment' and shows 'Environment is empty'. The R console at the bottom displays the R version (4.2.0) and the Xcode license acceptance process. The bottom right pane shows a sidebar with links to R resources, RStudio support, manuals, and reference materials.

Quiescent_germ_fungal_8Aug2018... | dada_phyloseq_JCexp.R | Intro_R_GradSkills_7oct2022.R

1 # Introduction to R
2 #By Carolina Sarmiento
3 RUSF - Grad Skills - Oct. 7, 2022
4
5 #####
6 # LET'S GET STARTED # -----
7 #####
8
9 # 1. Set your Working Directory (WD) -----
10 #You *ALWAYS* have to set your working directory in R. You will need and where you will need some of the
11 objects when you ask R to do something. So, you need to know where you are and where you want to go.
12 #A. Check current WD by running the following line
13 getwd()
14
15 #B. set your WD: Click on the drop-down menu "Session" -> Set Working Directory -> Choose Directory
16
17 # *** If you are using a package, you need to set the working directory to the package's working directory.
18 #####
19 setwd("~/Desktop/Grad_Skills/R/") # copy yours here!
20
21
22 # 2. Install a package -----
23
24 # A. Install the package
25 # Use first the "Test Package" button on the Packages tab. Look for the package of interest (dplyr) and make sure to check "install dependencies".
334 (Top) | R Script

R version 4.2.0 (2022-12-08) [64-bit]
Copyright (C) 2022 The R Foundation for Statistical Computing
Platform: x86_64-apple-darwin17.0 (64-bit)

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Type 'demo()' for some demos, 'help()' for on-line help, or
'help.start()' for an HTML browser interface to help.
Type 'q()' to quit R.

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This can occur if Xcode has been updated or reinstalled (e.g. as part of a macOS update).
Some features (e.g. Git / SVN) may be disabled.

Please run:

```
sudo xcodebuild -license accept
```

in a terminal to accept the Xcode license, and then restart RStudio.

>

Environment | History | Connections
R - Global Environment
Environment is empty

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Thanks
Technical papers



The image shows a screenshot of the RStudio IDE interface. The main editor window displays an R script with comments and code for setting the working directory and installing packages. The console window shows the R version (4.2.0) and the Xcode license acceptance process. A pink rectangular overlay is positioned on the right side of the image, containing a list of navigation links and resources. A large pink arrow points from the text 'Plots, Packages, and Help files' towards the pink overlay.

Plots, Packages, and Help files

R Resources

- Learning R Online
- CRAN Task Views
- R on StackOverflow
- Getting Help with R

RStudio

- RStudio IDE Support
- RStudio Community Forum
- RStudio Cheat Sheets
- RStudio Tip of the Day
- RStudio Packages
- RStudio Products

Manuals

- An Introduction to R
- Writing R Extensions
- R Data Import/Export

Reference

- Packages

Miscellaneous Material

- About R
- License
- NEWS
- Authors
- FAQ
- User Manuals
- Resources
- Thanks
- Technical papers

Search Engine & Keywords

A screenshot of the RStudio IDE. The main editor shows a script with R code for setting the working directory and installing packages. The console shows the R startup message. The Help pane on the right is highlighted with a pink border and contains a list of resources. A large pink arrow points from the text 'Plots, Packages, and Help files' to the Help pane.

Plots, Packages, and Help files

R Resources

- [Learning R Online](#)
- [CRAN Task Views](#)
- [R on StackOverflow](#)
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RStudio

- [RStudio IDE Support](#)
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- [RStudio Cheat Sheets](#)
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- [RStudio Packages](#)
- [RStudio Products](#)

Manuals

- [An Introduction to R](#)
- [Writing R Extensions](#)
- [R Data Import/Export](#)

Reference

- [The R Language Definition](#)
- [R Installation and Administration](#)
- [R Internals](#)

Packages

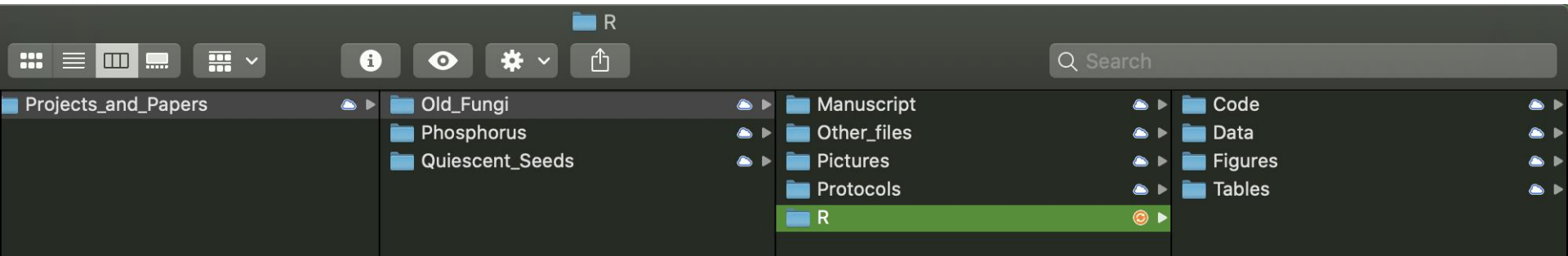
- [Search Engine & Keywords](#)

Miscellaneous Material

- [About R](#)
- [License](#)
- [NEWS](#)
- [Authors](#)
- [FAQ](#)
- [User Manuals](#)
- [Resources](#)
- [Thanks](#)
- [Technical papers](#)

Before starting:

1. Create a folder, dedicated to store the “R things” for your Grad Skills Class:



2. Move the file "Intro_R_GradSkills_10oct2025.R" to your R folder (Code)

3. Open  Studio®

4. Open the file "Intro_R_GradSkills_10oct2025.R"

1. Working Directory (WD)

Is a file path on your computer that sets the default location of any files you read into R, or save out of R:

If you ask R to import a dataset from a .csv file it will assume that the file is inside of your working directory.

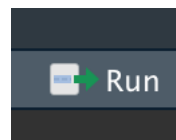
A. What is your current WD?

> getwd() *line 13*

Mac: command + Return

Win: ctrl + Enter

or click



at the top-right corner of
your text editor

*** Note: R will ignore all the lines preceded by #**

1. Working Directory (WD)

Is a file path on your computer that sets the default location of any files you read into R, or save out of R:

If you ask R to import a dataset from a .csv file it will assume that the file is inside of your working directory.

A. What is your current WD?

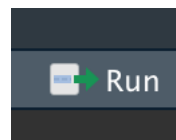
`> getwd( )` *line 13*

*Read all the text in the line
and run each line at a time*

Mac: command + Return

Win: ctrl + Enter

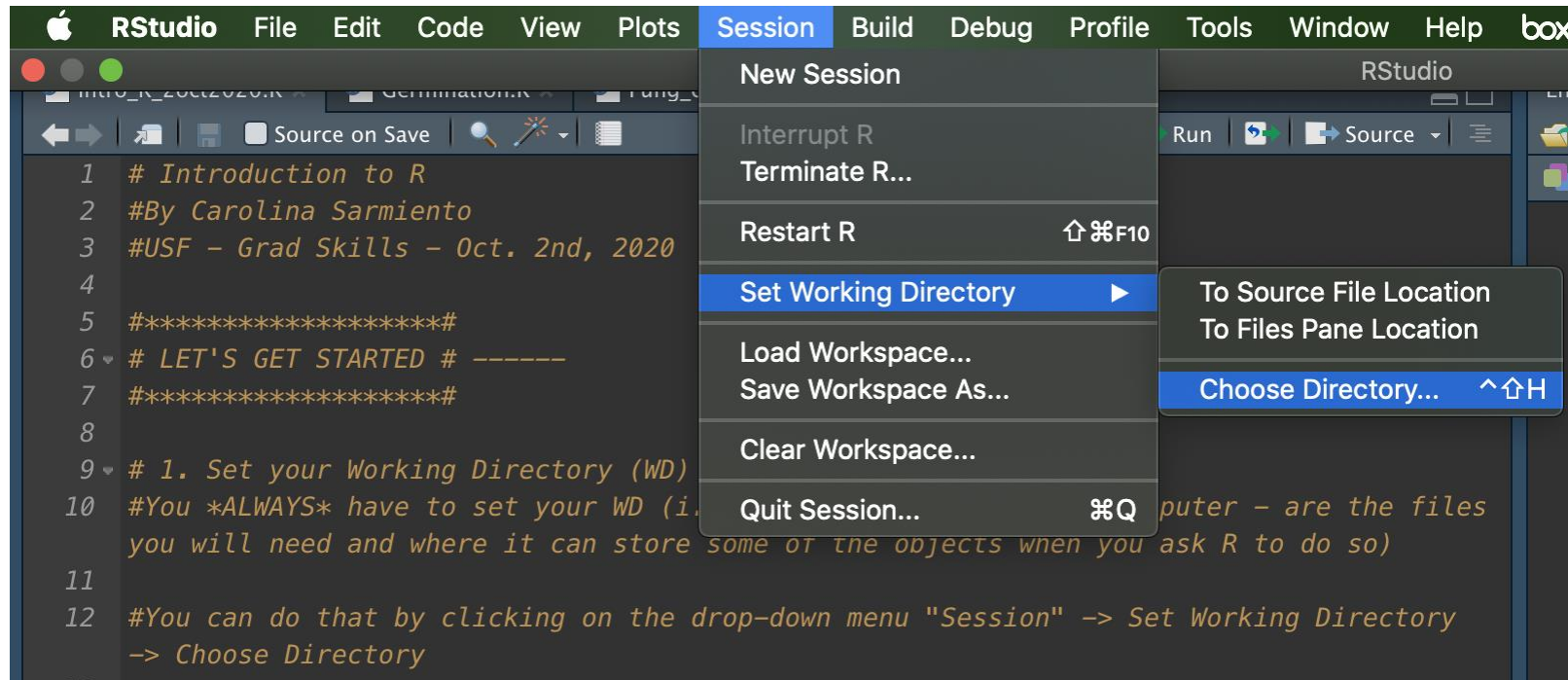
or click



at the top-right corner of
your text editor

*** Note: R will ignore all the lines preceded by #**

1. Working Directory (WD)



B. Set your WD:

Or run the line:

`>setwd("path")` ---- > Check your console

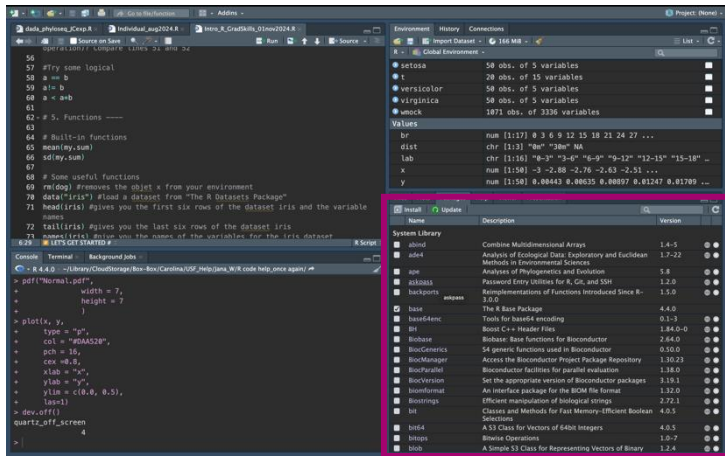
Here is mine:

`> setwd("~/Desktop/Grad_Skills/R")`

line 19

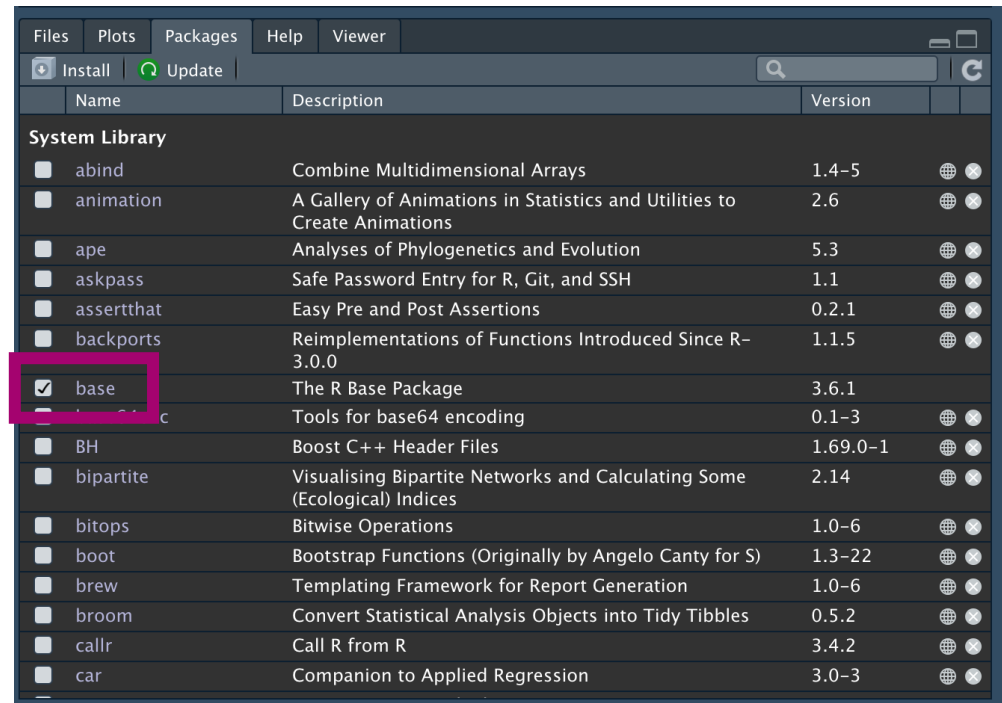
2. Install a package

Let's take a moment to check your **Packages** tab on the bottom-right



The screenshot shows the RStudio interface. The script editor on the left contains R code for creating variables, defining functions, and loading the 'iris' dataset. The Environment pane on the right shows the current environment with variables like 'br', 'dist', 'lab', 'x', and 'y'. The Packages pane at the bottom right shows a list of installed and available packages, with 'base' highlighted.

```
56 # Try some logical
57 a = b
58 a != b
59 a < b
60 a <= b
61 # 5. Functions
62
63 # Built-in functions
64 mean(my.sum)
65 my.sum
66 # Some useful functions
67 # ref() removes the object x from your environment
68 data("iris") # load a dataset from "The R Datasets Package"
69 head(iris) # gives you the first six rows of the dataset iris and the variable
70 # names
71 tail(iris) # gives you the last six rows of the dataset iris
72 # ref() removes the object x from your environment
73 # ref() removes the object x from your environment
74 # ref() removes the object x from your environment
75 # ref() removes the object x from your environment
76 # ref() removes the object x from your environment
77 # ref() removes the object x from your environment
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96 # ref() removes the object x from your environment
97 # ref() removes the object x from your environment
98 # ref() removes the object x from your environment
99 # ref() removes the object x from your environment
100 # ref() removes the object x from your environment
```

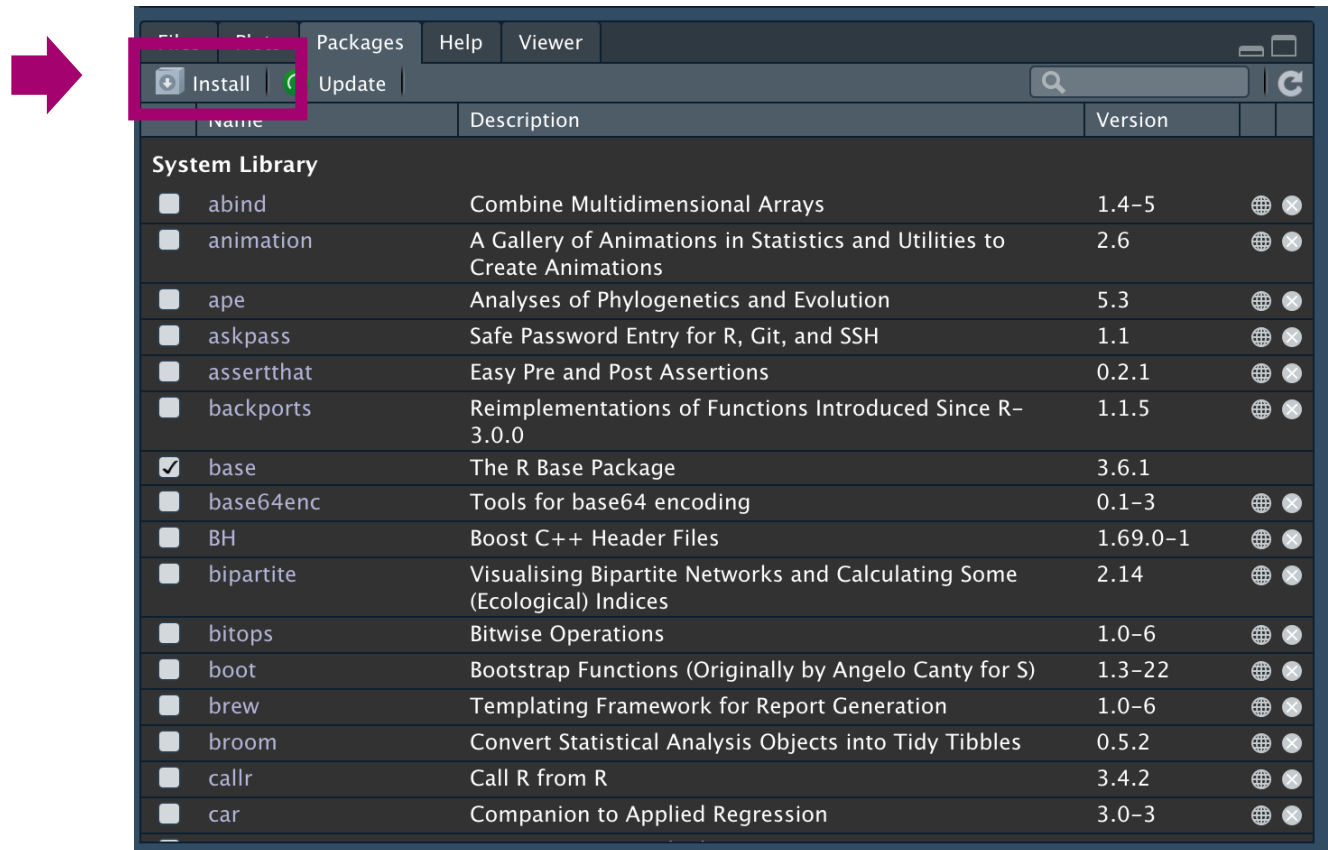


The screenshot shows the RStudio Packages tab. It displays a list of installed and available packages. The 'base' package is highlighted with a red box. The table below shows the details of the packages.

Name	Description	Version
System Library		
☐ abind	Combine Multidimensional Arrays	1.4-5
☐ animation	A Gallery of Animations in Statistics and Utilities to Create Animations	2.6
☐ ape	Analyses of Phylogenetics and Evolution	5.3
☐ askpass	Safe Password Entry for R, Git, and SSH	1.1
☐ assertthat	Easy Pre and Post Assertions	0.2.1
☐ backports	Reimplementations of Functions Introduced Since R-3.0.0	1.1.5
☒ base	The R Base Package	3.6.1
☐ base64enc	Tools for base64 encoding	0.1-3
☐ BH	Boost C++ Header Files	1.69.0-1
☐ bipartite	Visualising Bipartite Networks and Calculating Some (Ecological) Indices	2.14
☐ bitops	Bitwise Operations	1.0-6
☐ boot	Bootstrap Functions (Originally by Angelo Canty for S)	1.3-22
☐ broom	Templating Framework for Report Generation	1.0-6
☐ callr	Convert Statistical Analysis Objects into Tidy Tibbles	0.5.2
☐ callr	Call R from R	3.4.2
☐ car	Companion to Applied Regression	3.0-3

2. Install a package

We will install the package **dslabs**



2. Install a package

We will install the package **dslabs**

From CRAN (default)



Path to the folder where it will be stored in your computer



Install Packages

Install from: [? Configuring Repositories](#)
Repository (CRAN)

Packages (separate multiple with space or comma):
dslabs

Library:
/Library/Frameworks/R.framework/Versions/3.6/Resources/library [C]

☒ Install dependencies

Install Cancel



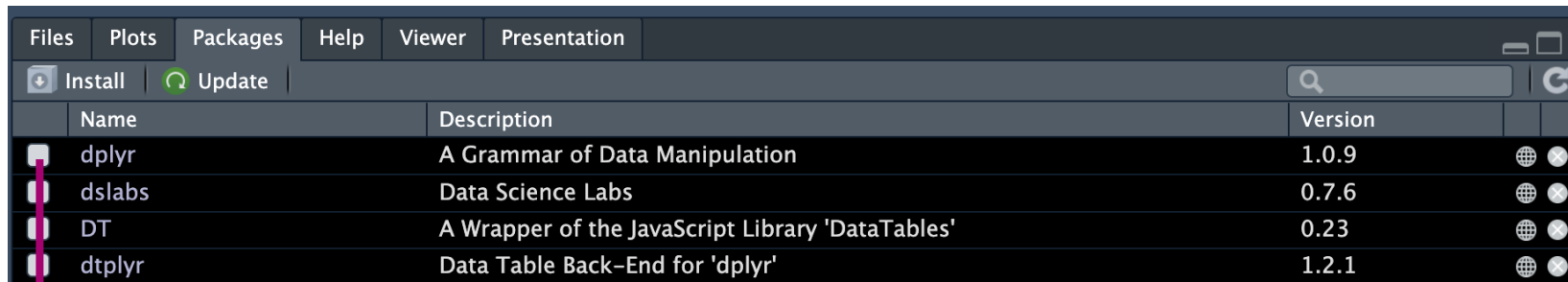
Make sure this box is checked

What happened in the console?

```
> install.packages("dslabs")
```

2. Install a package

Now that we are here (Packages tab):



Files	Plots	Packages	Help	Viewer	Presentation	
+	Install	Update				
	Name	Description	Version			
☒	dplyr	A Grammar of Data Manipulation	1.0.9			
☒	dslabs	Data Science Labs	0.7.6			
☒	DT	A Wrapper of the JavaScript Library 'DataTables'	0.23			
☒	dtplyr	Data Table Back-End for 'dplyr'	1.2.1			

The package is stored in your "library" but to access it, you need to load the package (i.e., check that box!)

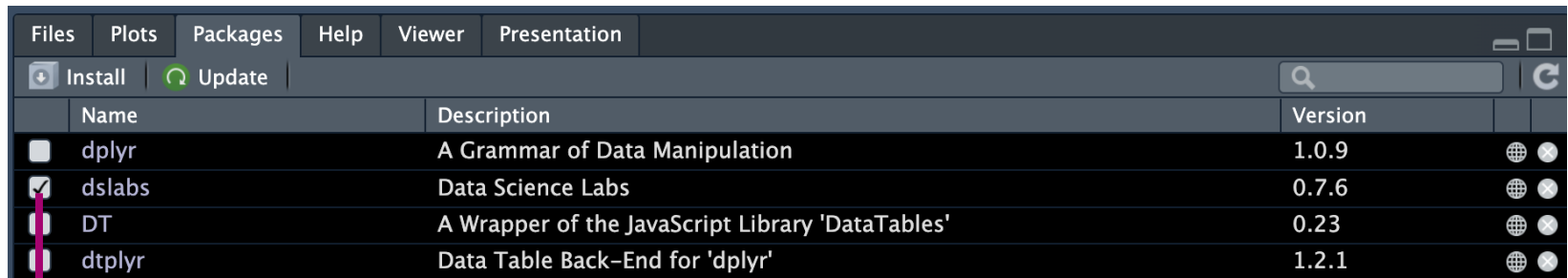
or

run the line with the function *library()*

> **library(dslabs)**

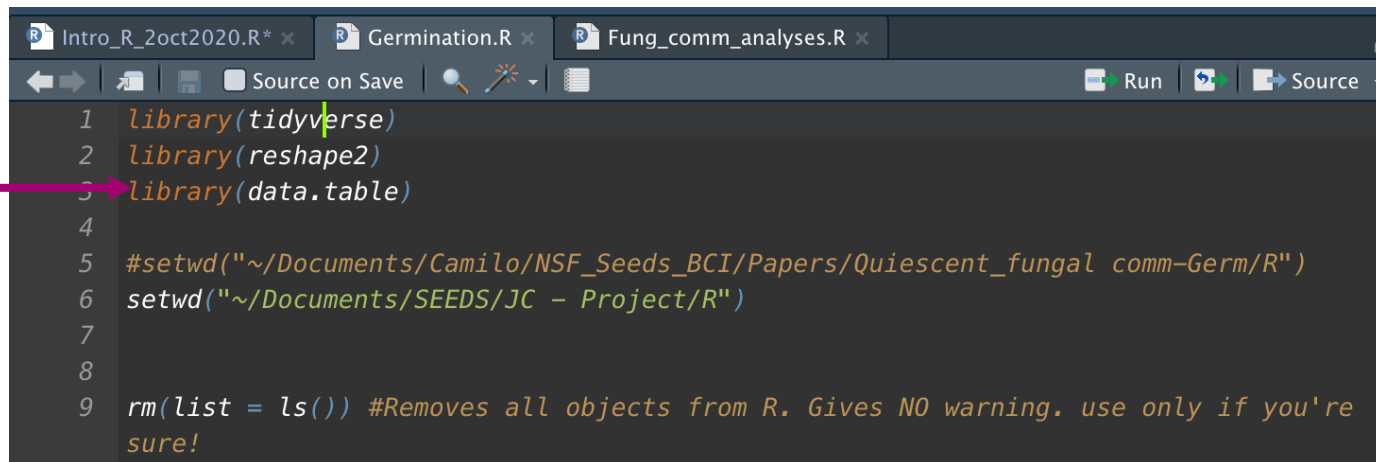
Tip: copy this line at the top of your script, so you don't forget to load all the packages you need for your code to run

2. Install a package



The RStudio Packages pane displays a list of installed and available packages. The 'Install' button is highlighted in green. A red arrow points from the 'library(data.table)' line in the script editor below to the 'DT' package entry in this pane.

	Name	Description	Version		
☐	dplyr	A Grammar of Data Manipulation	1.0.9		
☒	dslabs	Data Science Labs	0.7.6		
☐	DT	A Wrapper of the JavaScript Library 'DataTables'	0.23		
☐	dtplyr	Data Table Back-End for 'dplyr'	1.2.1		



The RStudio script editor shows the following R code. A red arrow points from the 'library(data.table)' line in the script editor to the 'DT' package entry in the Packages pane above.

```
1 library(tidyverse)
2 library(reshape2)
3 library(data.table)
4
5 #setwd("~/Documents/Camilo/NSF_Seeds_BCI/Papers/Quiescent_fungal_comm-Germ/R")
6 setwd("~/Documents/SEEDS/JC - Project/R")
7
8
9 rm(list = ls()) #Removes all objects from R. Gives NO warning. use only if you're sure!
```

2. Install a package

Now that we are here:

☐	vectors		
☐	dplyr	A Grammar of Data Manipulation	0.8.3
☐	dslabs	Data Science Labs	0.7.3
☐	DT	A Wrapper of the JavaScript Library 'DataTables'	0.9

If you click in the package name:

Full name

Description file for the package

List of “objects” contained in the package
(usually functions and datasets) and their own
help files.

The screenshot shows the R Data Science Labs package documentation page for 'dslabs' version 0.7.3. The page has a dark theme with a blue R logo. A pink box highlights the 'Packages' menu in the top navigation bar. A pink arrow points from the 'Full name' label to the 'Data Science Labs' header. Another pink arrow points from the 'Description file for the package' label to the 'DESCRIPTION file.' link. A pink bracket on the left groups the 'Help Pages' list, and a pink bracket on the right groups the corresponding descriptions.

admissions	Gender bias among graduate school admissions to UC Berkeley.
brca	Breast Cancer Wisconsin Diagnostic Dataset from UCI Machine Learning Repository
brexit_polls	Brexit Poll Data
death_prob	2015 US Period Life Table
divorce_margarine	Divorce rate and margarine consumption data
ds_theme_set	dslabs theme set
gapminder	Gapminder Data
greenhouse_gases	Greenhouse gas concentrations over 2000 years
heights	Self-Reported Heights
historic_co2	Atmospheric carbon dioxide concentration over 800,000 years
mnist_27	Useful example for illustrating machine learning algorithms based on MNIST data
movielens	Movie ratings

So far...

- Using a script (**as opposed to typing your instructions directly on the console**) is highly recommended
-  Studio® makes easier some common tasks
- **#** is useful to comment your code
 - R “ignores” the text preceded by **#**
 - And I strongly suggest to use it (as much as you can!)
- Rstudio’s text editor recognize some elements and color them differently (**syntax-highlighting editor**) which is very useful

So far...

- If your console panel is active: Arrow up returns previously entered code

- R is case sensitive: “Anova” is different from “anova”

- Quotation marks must not be directional (“NO”; "YES")

Do not use Microsoft Word as text editor!

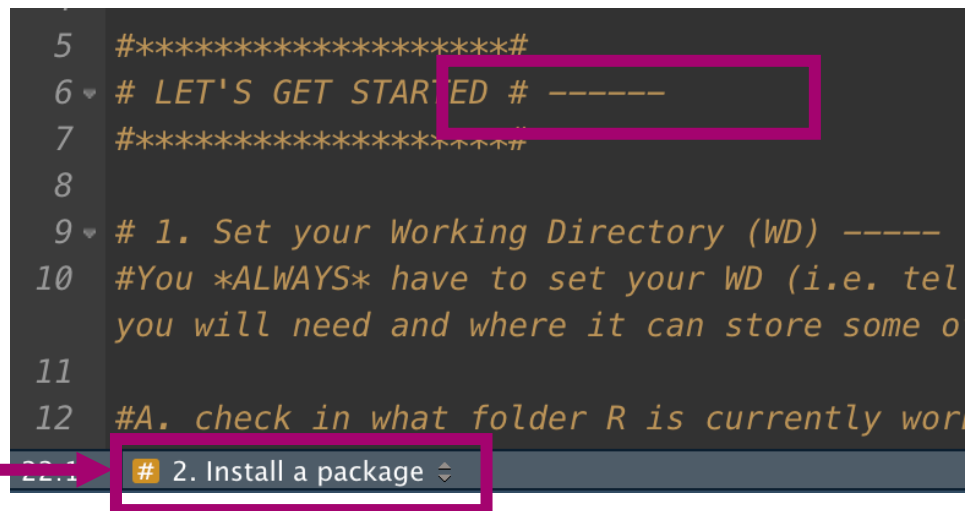
- Click  at the bottom of your text editor:

```
5 *****#
6 # LET'S GET STARTED # -----
7 *****#
8
9 # 1. Set your Working Directory (WD) -----
10 #You *ALWAYS* have to set your WD (i.e. tell
    you will need and where it can store some o
11
12 #A. check in what folder R is currently worki
22:1 # 2. Install a package
```

So far...

- If your console panel is active: Arrow up returns previously entered code
- R is case sensitive: “Anova” is different from “anova”
- Quotation marks must not be directional (“NO”; "YES")
Do not use Microsoft Word as text editor!
- Click  at the bottom of your text editor:

It indexes your script by using
“----” Four or more dashes or #



```
5 *****#
6 # LET'S GET STARTED # ----
7 *****#
8
9 # 1. Set your Working Directory (WD) ----
10 #You *ALWAYS* have to set your WD (i.e. tel
    you will need and where it can store some o
11
12 #A. check in what folder R is currently wor
13
14 # 2. Install a package #
```

3. Objects

R is an object-oriented language:

Variables, data, functions, results, etc., are stored in the active memory of the computer in the form of *objects*, which have a *name*.

`name <- object`

(To assign an object (value, data, function) to a name use `<-`)

Mac: option + -

Win: alt + -

3. Objects

R is an object-oriented language:

Variables, data, functions, results, etc., are stored in the active memory of the computer in the form of *objects*, which have a *name*.

name ← object

```
> a <- 15
```

```
> b <- c(1:10)
```

```
> c <- "My dog's name is Kona"
```

*Note that if you use the same name, R will replace the object without asking

*** e.g., the values 1 to 10 are stored in the object named “b”, to access it, type b in your console + return/enter ***

lines 32 - 49

*Rstudio will let you know when a line is “evidently” wrong (line 45)

3. Objects

R is an object-oriented language:

Variables, data, functions, results, etc., are stored in the active memory of the computer in the form of *objects*, which have a *name*.

name $\leftarrow$ object

```
> a <- 15
```

```
> b <- c(1:10)
```

```
> c <- "My dog's name is Kona"
```

The user can create and modify these objects with *operators* and *functions*

```
> a * b
```

```
> my.sum <- a + b
```

4. Operators

□ Arithmetic

Operator	Description
+	addition
-	subtraction
*	multiplication
/	division
^ or **	exponentiation

> a * b

□ Logical

Operator	Description
<	less than
<=	less than or equal to
>	greater than
>=	greater than or equal to
==	exactly equal to
!=	not equal to
!x	Not x
x y	x OR y
x & y	x AND y

> a == b

lines 52 - 64

5. Functions

Almost everything in R is done through functions

`function.name (arguments)`

- **Built-in Functions** (they can be numeric, character, statistical, probability, etc.)

Function	Description
<code>abs(x)</code>	absolute value
<code>sqrt(x)</code>	square root
<code>round(x, digits=n)</code>	<code>round(3.475, digits=2)</code> is 3.48
<code>log(x)</code>	natural logarithm
<code>log10(x)</code>	common logarithm
<code>exp(x)</code>	e^x
<code>sum(x)</code>	sum
<code>min(x)</code>	minimum
<code>max(x)</code>	maximum

We already know some others:

`getwd( )`

`setwd( )`

`install.packages("dslabs")`

`library(dslabs)`

`c( )` *concatenate, useful to create vectors

<code>mean(x, na.rm=FALSE)</code>	mean of object x
<code>sd(x)</code>	standard deviation of object(x). also look at <code>var(x)</code> for variance and <code>mad(x)</code> for median absolute deviation.
<code>median(x)</code>	median

5. Functions

Almost everything in R is done through functions

- **User-written functions:** One of the great strengths of R is the user's ability to add functions.

```
myfunction <- function(arg1, arg2, ... ){  
  statements  
  return(object)  
}
```

```
> myfunction1 <- function(x, y=0){  
  print (x)  
  print (y)  
  return (sum ( (x - y) ^ 2 ) )  
}
```



```
> myfunction1(10)  
[1] 10  
[1] 0  
[1] 100  
> myfunction1(10,5)  
[1] 10  
[1] 5  
[1] 25
```

lines 66 - 103

6. Object properties

- **Mode:** how objects are stored in R
character, numeric, logical, complex, & function

```
> mode(iris$Sepal.Length)  
[1] "numeric"
```
- **Class:** how objects are treated by functions
character, factor, numeric, matrix, data.frame, etc.

```
> class(iris$Species)  
[1] "factor"
```
- **Length:** Number of elements of an object

```
> length(b)  
[1] 10
```

7. Data structure

□ **Vector:** list of elements of the same type (mode) 1-dimensional

```
> e <- c(TRUE, TRUE, FALSE, FALSE, FALSE)
> str(e)
logi [1:5] TRUE TRUE FALSE FALSE FALSE
```

□ **Matrix:** elements in rows and columns, same type 2-dimensional

```
> b <- c(1:10)
> d <- c(1 1:20)
> g <- cbind(b,d) *function cbind() joins the two objects as columns
```

□ **Data Frame:** like a matrix (rows,columns) but different columns can have different modes 2-dimensional

```
> x <- c(1:10)
> y <- rep(TRUE,10)
> z <- c("a", "b", "c", "d", "e", "f", "g", "h", "i", "j")
> mydf <- data.frame(x,y,z)
```

lines 124 - 138

8. Accessing the values of an object

The indexing system

Matrices and Data frames have a similar structure

[rows , columns]

```
> A <- matrix(rnorm(100),10,10) *check >help("rnorm")
```

>A[3,]	shows 3 rd row and all columns
>A[,4]	shows all rows and the 4 th column
>A[3,4]	shows element in row 3, column 4
>A[c(1,9),]	shows rows 1 and 9, all columns
>A[,-1]	shows all of A, except column 1
>A[-c(1,9),]	shows all of A, except rows 1 and 9
>A[1:10,5:8]	shows first 10 rows and columns 5 to 8
>A[A[,1]>0,]	shows only rows where first column is a positive number
>A[10:1,]	shows first 10 rows of A, in opposite order

8. Accessing the values of an object

The indexing system

Matrices and Data frames have a similar structure

[rows , columns]

A data frame has **column names that can be used to access its elements**

`> mydf <- iris` To access one column by its name use `$`
(e.g., `mydf$Species`)

A. Show all the rows that meet the condition 'Species = setosa':

```
>mydf [mydf$Species == "setosa", ]
```

B. Show all the rows with sepal lengths > 6 cm and sepal widths < 4 cm:

```
>mydf [mydf$Sepal.Length > 6 & mydf$Sepal.Width < 4, ]
```

lines 141 - 174

9. Import a dataset

There are several ways of importing data into R:

A. The one that I use and recommend:

```
>mydata <- read.csv("Data/Iris_data.csv", header = TRUE)
```

imports a .csv file stored in the Data sub folder of my WD and recognizes the first row as column names

B.

```
>mydata <- read.table("Data/Iris_data.txt", sep= "\t", h = T)
```

Imports a text file and recognizes the first row as column names

Also

```
>read.table("Data/Iris_data.csv", sep = ",", header =TRUE)
```

C.

```
>mydata <- read.csv(file.choose( ) )
```

Imports a .csv file from anywhere in your computer

lines 176 - 186

9. Import a dataset

Common problems when importing data:

- R doesn't recognize the path you gave or can't find the file:

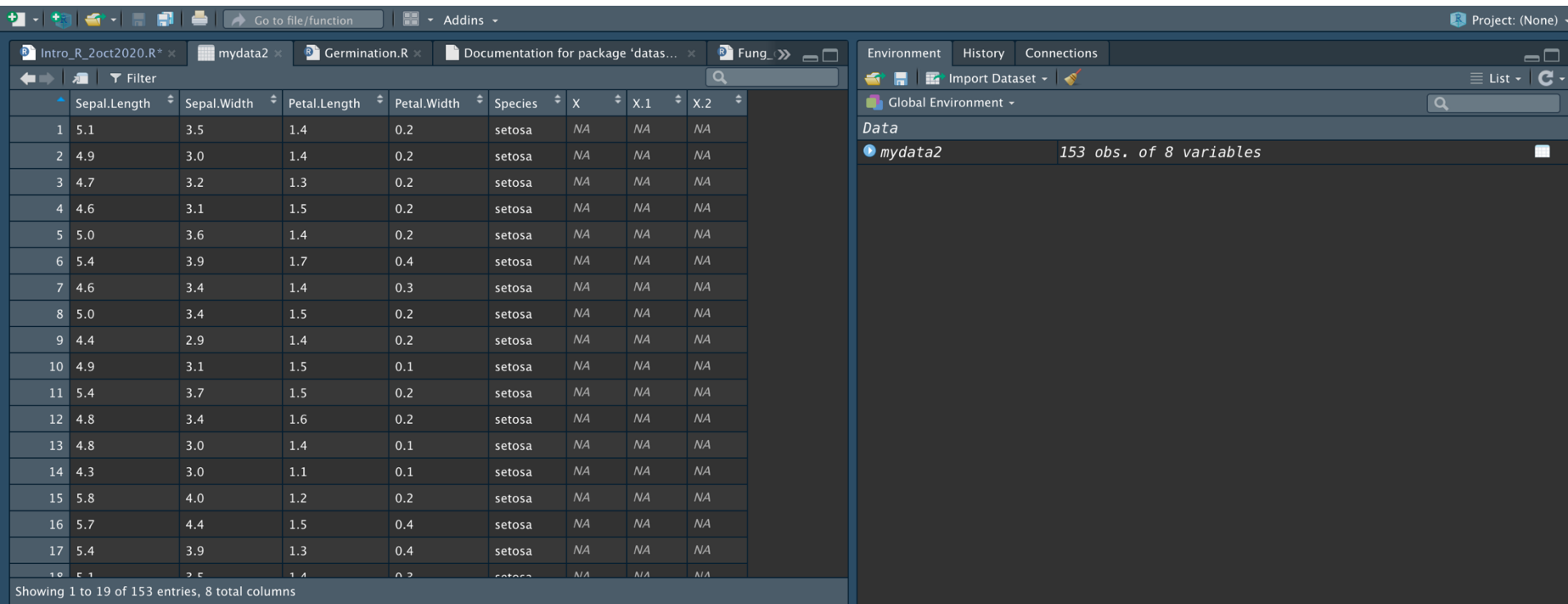
```
> mydata <- read.table("Data/Iris_data.txt", sep= "\t", h = T)
Error in file(file, "rt") : cannot open the connection

In addition: Warning message:
In file(file, "rt") :
  cannot open file 'Data/Iris_data.txt': No such file or directory
```

9. Import a dataset

Common problems when importing data:

- Your data set has spaces (that you can't notice in Excel!)



The screenshot shows the RStudio interface with the following components:

- Top Panel:** Shows the file explorer with tabs for 'Intro_R_2oct2020.R', 'mydata2', 'Germination.R', 'Documentation for package 'datis...', and 'Fung_'. The 'mydata2' tab is active.
- Left Panel:** Displays a data frame with 19 rows and 8 columns. The columns are Sepal.Length, Sepal.Width, Petal.Length, Petal.Width, Species, X, X.1, and X.2. The data is sorted by Sepal.Length in ascending order.
- Right Panel:** Shows the 'Environment' pane with a search bar and a list of objects. The object 'mydata2' is listed with the description '153 obs. of 8 variables'.

	Sepal.Length	Sepal.Width	Petal.Length	Petal.Width	Species	X	X.1	X.2
1	5.1	3.5	1.4	0.2	setosa	NA	NA	NA
2	4.9	3.0	1.4	0.2	setosa	NA	NA	NA
3	4.7	3.2	1.3	0.2	setosa	NA	NA	NA
4	4.6	3.1	1.5	0.2	setosa	NA	NA	NA
5	5.0	3.6	1.4	0.2	setosa	NA	NA	NA
6	5.4	3.9	1.7	0.4	setosa	NA	NA	NA
7	4.6	3.4	1.4	0.3	setosa	NA	NA	NA
8	5.0	3.4	1.5	0.2	setosa	NA	NA	NA
9	4.4	2.9	1.4	0.2	setosa	NA	NA	NA
10	4.9	3.1	1.5	0.1	setosa	NA	NA	NA
11	5.4	3.7	1.5	0.2	setosa	NA	NA	NA
12	4.8	3.4	1.6	0.2	setosa	NA	NA	NA
13	4.8	3.0	1.4	0.1	setosa	NA	NA	NA
14	4.3	3.0	1.1	0.1	setosa	NA	NA	NA
15	5.8	4.0	1.2	0.2	setosa	NA	NA	NA
16	5.7	4.4	1.5	0.4	setosa	NA	NA	NA
17	5.4	3.9	1.3	0.4	setosa	NA	NA	NA
18	5.1	3.5	1.4	0.2	setosa	NA	NA	NA

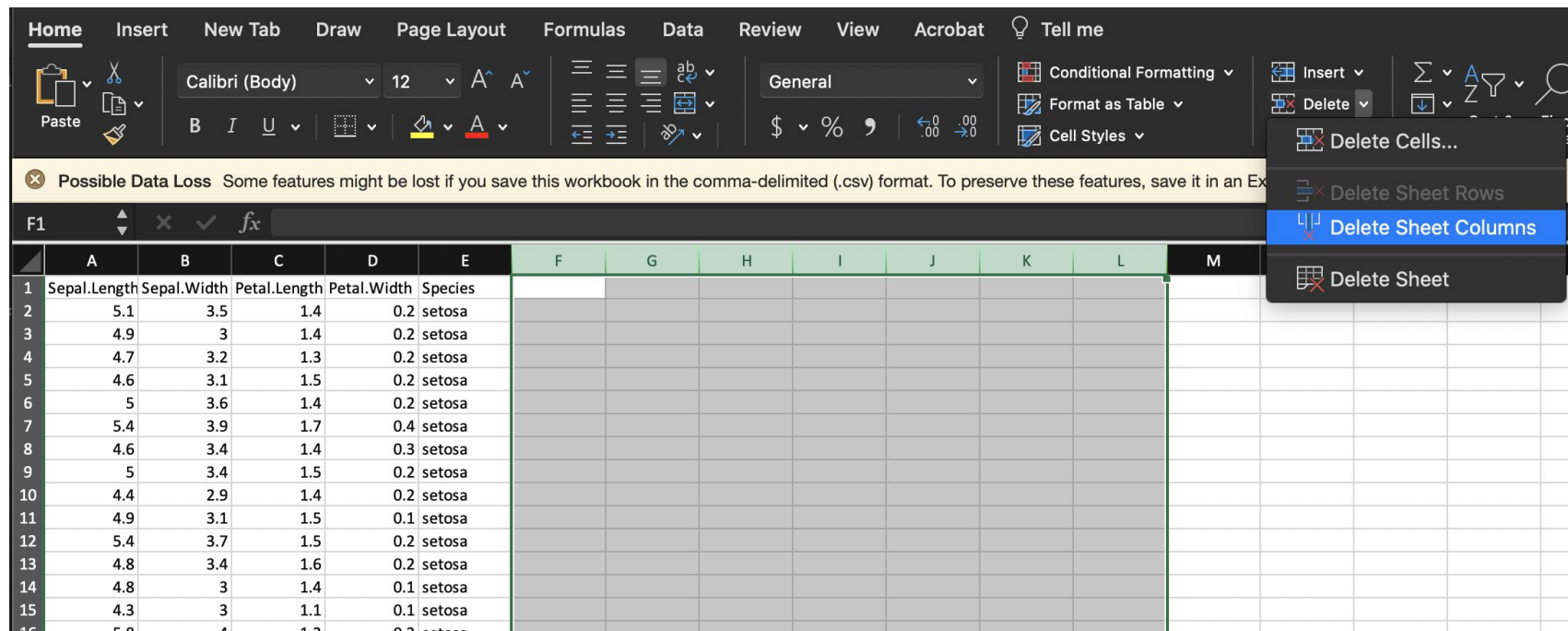
Showing 1 to 19 of 153 entries, 8 total columns

9. Import a dataset

Common problems when importing data:

- Your data set has spaces (that you can't notice in Excel!)

To fix this, I suggest to open your .csv file in Excel, select several columns and delete them

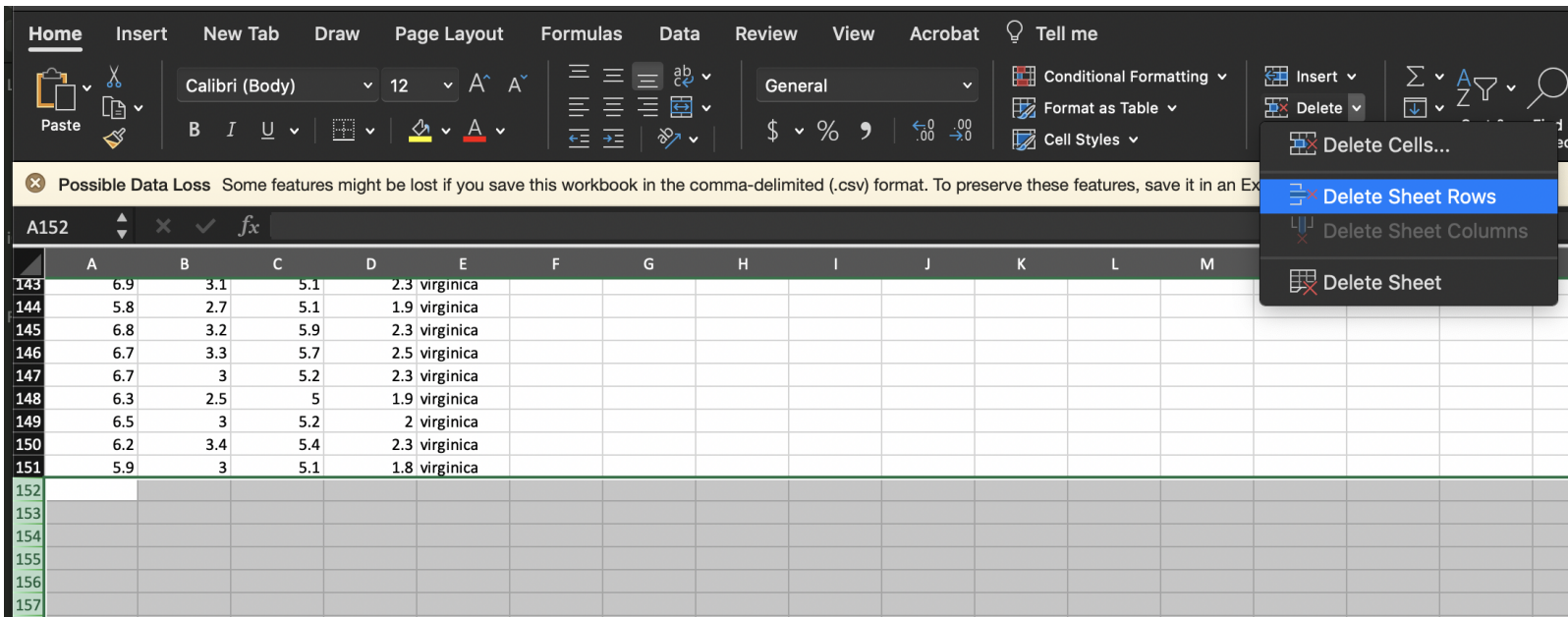


9. Import a dataset

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9. Import a dataset

Common problems when importing data:

- Your data set has empty cells, or “weird” things that don’t let you get what you want:

The screenshot shows the RStudio environment with a dataset named 'mydata2' loaded. The dataset has 150 observations and 8 variables. The variables are: Sepal.Length (Factor), Sepal.Width (num), Petal.Length (num), Petal.Width (num), Species (Factor), X (logi), X.1 (logi), and X.2 (logi). The data is displayed in a table with columns: Sepal.Length, Sepal.Width, Petal.Length, Petal.Width, Species, X, X.1, and X.2. The first 7 rows are visible, with the 5th row highlighted in pink. The 5th row has a value of 'x' in the Sepal.Length column, which is an invalid factor level. The console shows the command `View(mydata2)` and the error message: `Error in Summary.factor(c(9L, 7L, 5L, 4L, 36L, 12L, 4L, 8L, 2L, 7L, 12L, : 'range' not meaningful for factors)`. The Environment pane shows the dataset 'mydata2' with 150 observations and 8 variables. The Files pane shows the RStudio interface with the 'read.table' function highlighted.

	Sepal.Length	Sepal.Width	Petal.Length	Petal.Width	Species	X	X.1	X.2
1	5.1	3.5	1.4	0.2	setosa	NA	NA	NA
2	4.9	3.0	1.4	0.2	setosa	NA	NA	NA
3	4.7	3.2	1.3	0.2	setosa	NA	NA	NA
4	4.6	3.1	1.5	0.2	setosa	NA	NA	NA
5	x	3.6	1.4	0.2	setosa	NA	NA	NA
6	5.4	3.9	1.7	0.4	setosa	NA	NA	NA
7	4.6	3.4	1.4	0.3	setosa	NA	NA	NA

```
> View(mydata2)
> range(mydata2$Sepal.Length)
Error in Summary.factor(c(9L, 7L, 5L, 4L, 36L, 12L, 4L, 8L, 2L, 7L, 12L, :
  'range' not meaningful for factors
>
```

Depending on the size of your dataset, you can check carefully on excel or a text editor or sometimes you must clean your dataset in R

10. Export a dataset

Sometimes you need to export your objects.

For data you can use:

```
write.table(object, "path", sep=",")  
write.csv(object, "path/file.name")  
> write.csv(mydata, "Data/New_iris.csv")
```



This will save a new .csv file inside the folder “Data” on my WD

lines 189 - 194

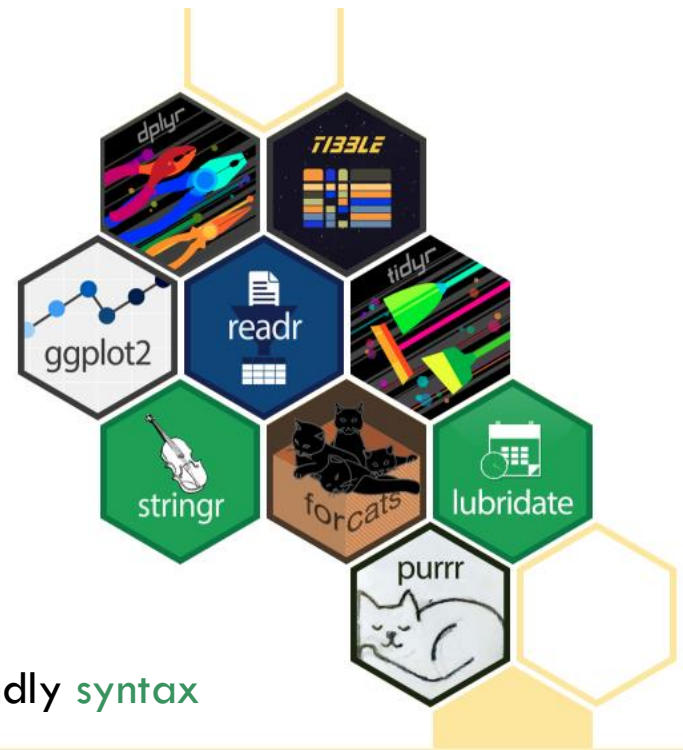
The *tidyverse*

**A collection of R packages for data science.
They share a design philosophy, grammar, and
data structures.**

Tools for:

- data import
- data wrangling
- visualization
- modeling

Makes code more readable and consistent
through features like data **pipelines** and a friendly **syntax**



The *tidyverse*

A different syntax: using the pipe operator (`%>%` or `|>`) to chain together a series of functions



Clear and readable sequence of operations that flow from one step to the next

Instead of nesting function calls: `function3(function2(function1(data)))`,

chain them using the pipe operator - more readable and intuitive

data `%>%`

`function1( ) %>%`

`function2( ) %>%`

`function3( )`

The *tidyverse*

A different syntax: using the pipe operator (`%>%` or `|>`) to chain together a series of functions



Creates a clear and readable sequence of operations that flow from one step to the next

In base R:

```
cars.6.cyl <- mtcars[mtcars$cyl == 6,]  
my.df <- cars.6.cyl[, c("mpg", "hp")]  
print(my.df)
```

Using the *tidyverse*:

```
my.df <- mtcars %>%  
  filter(cyl == 6) %>%  
  select(mpg, hp)
```

dplyr and *ggplot2*

dplyr: Data manipulation – set of *verbs* that help you solve the most common challenges

- ☐ **mutate()**: add new variables
- ☐ **select()**: select variables (columns)
- ☐ **filter()**: select cases (rows)
- ☐ **summarise()**: reduce multiple values to a single summary
- ☐ **arrange()**: order rows

11. R Graphics

The graphical features in R are one of its major strengths

To get an idea what is possible with R run:

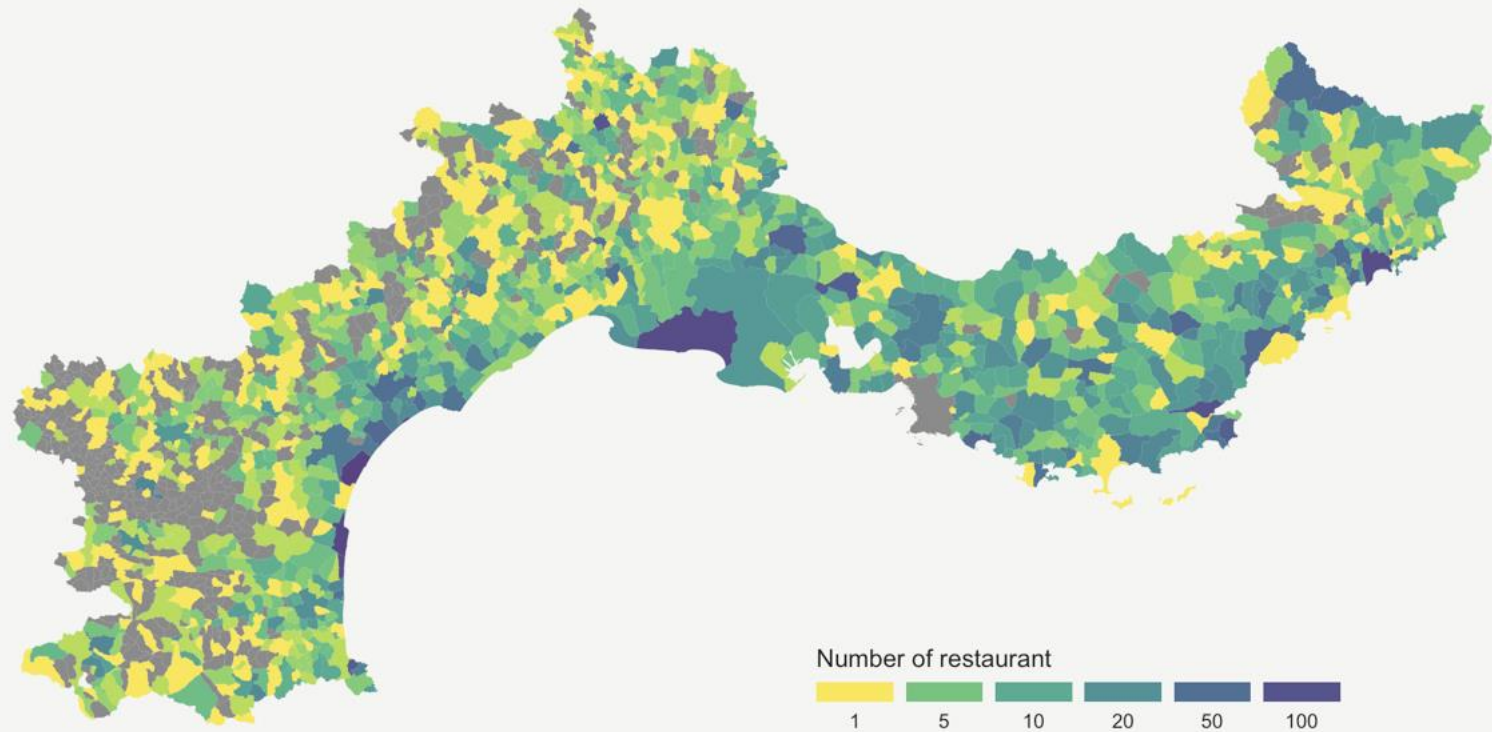
```
>demo(graphics)
```

or visit:

<https://www.r-graph-gallery.com>

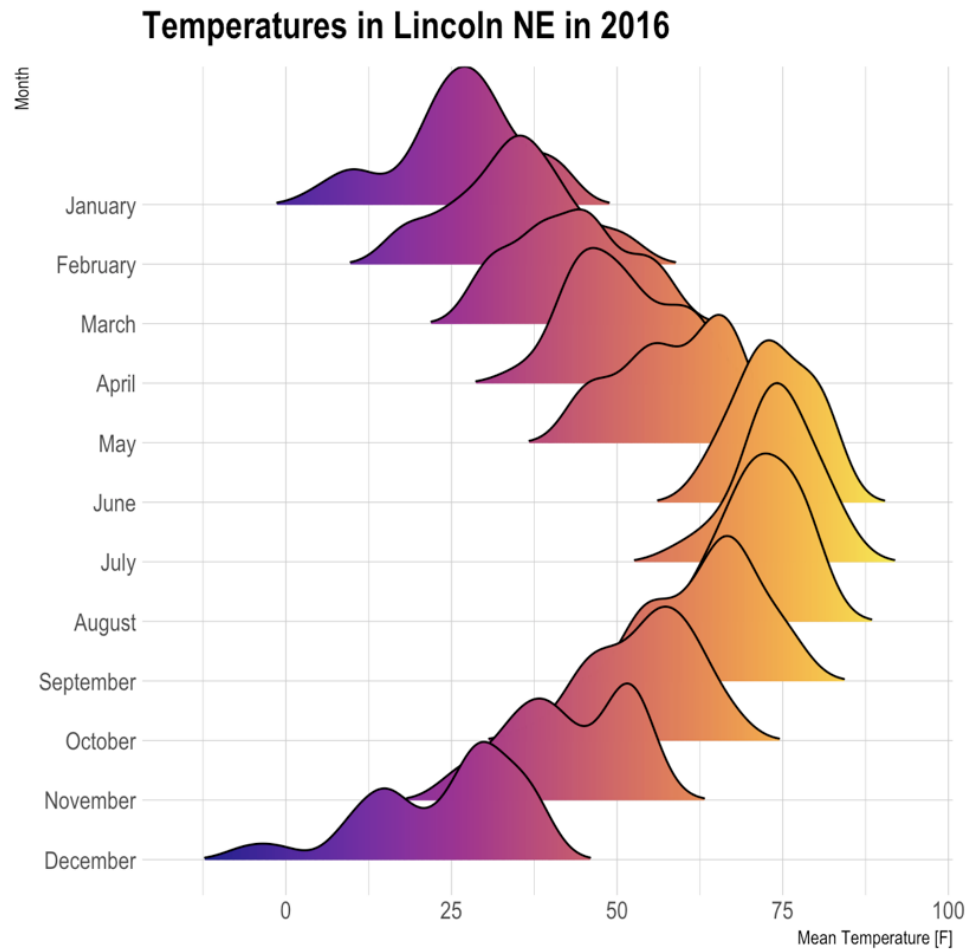
11. R Graphics

South of France Restaurant concentration
Number of restaurant per city district



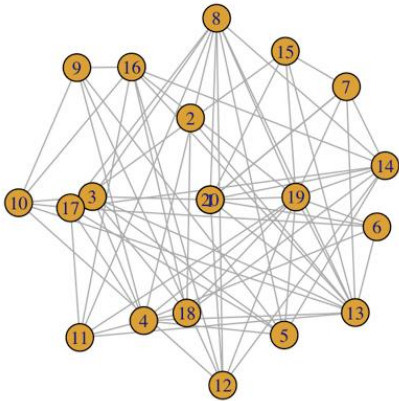
Data: INSEE | Creation: Yan Holtz | r-graph-gallery.com

11. R Graphics

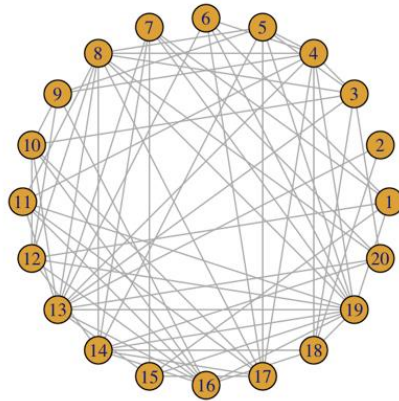


11. R Graphics

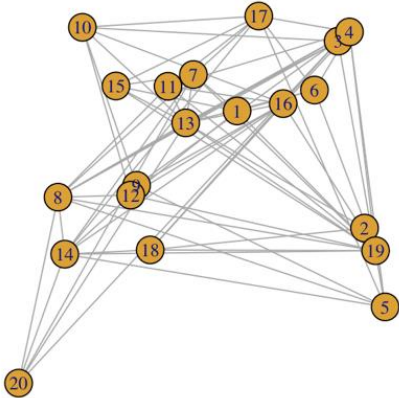
sphere



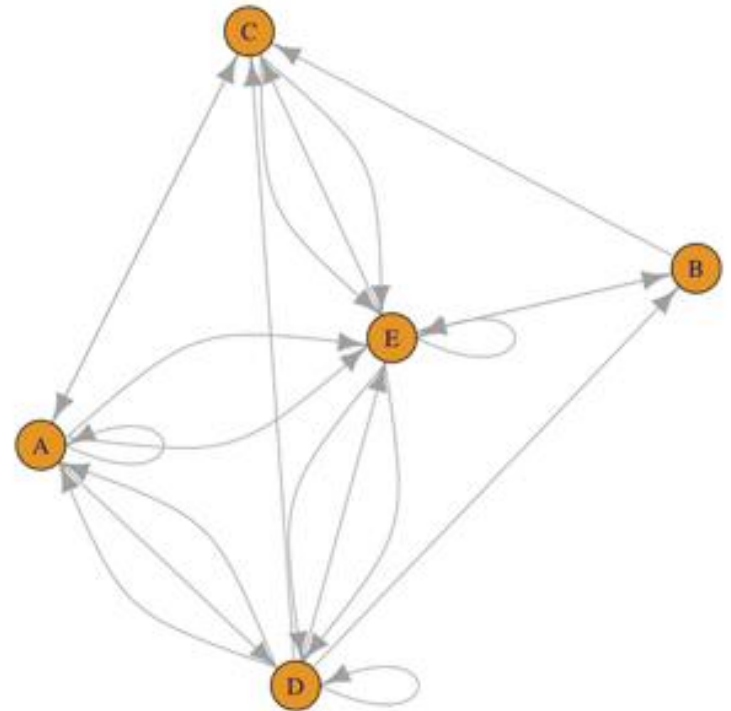
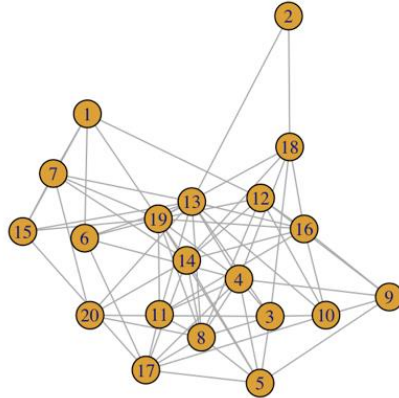
circle



random



fruchterman.reingold



11. R Graphics

Generic “plot “ function – graphic type depends on object type:

plot(x,y)

if x, y are numeric vectors creates a scatterplot

plot(x)

if x is a factor creates a barplot

plot(x,y)

if x is a factor and y a numeric vector creates a boxplot of y at each level of x

plot(x)

if x is a data frame, creates distributional plots for all components

12. Getting help

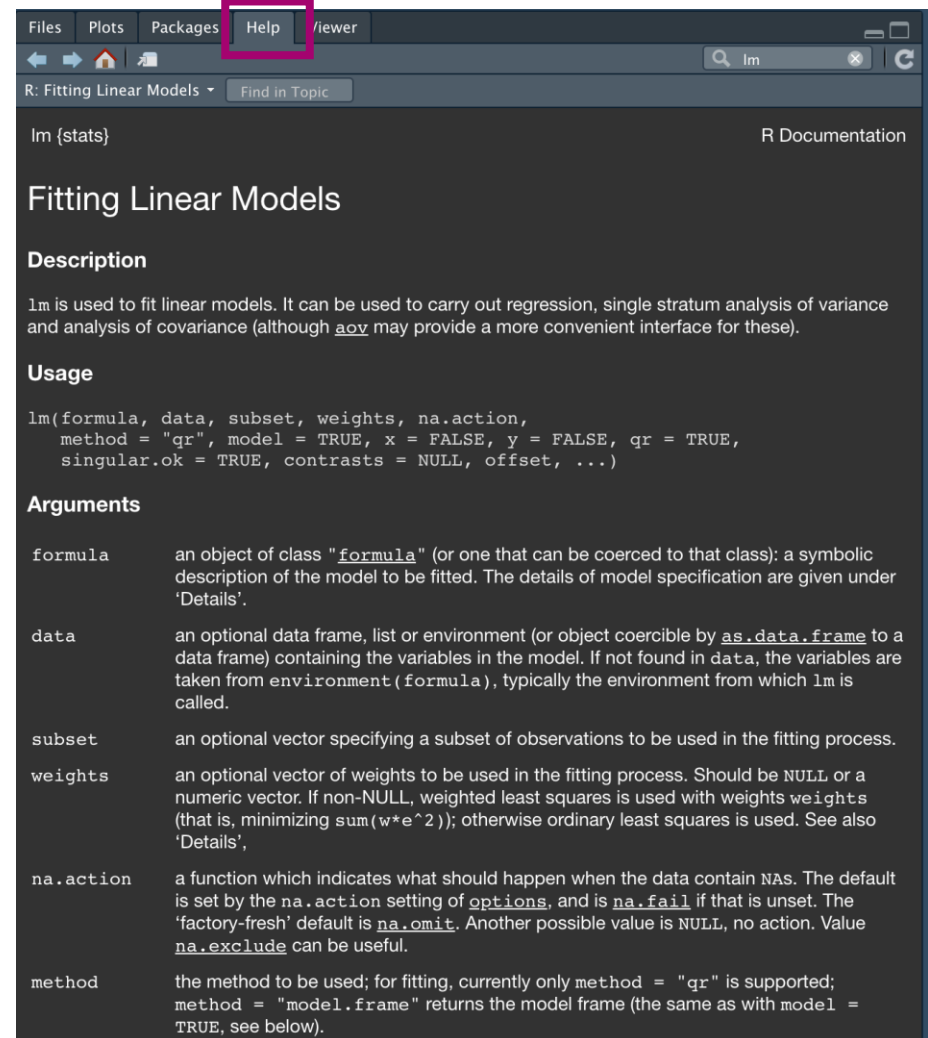
□ To get the documentation file for a function:

```
> help("anova")
```

```
> help(anova)
```

```
> ?anova
```

□ Easy in RStudio: just type in the function/package name in the 'help' tab



12. Getting help

- If this fails, try searching the R website or www.rseek.org



Created and maintained by [Sasha Goodman](#).
Serving the R community since 2007. Version 2.0.

[Privacy Policy](#)

[Download and Install R](#)

- Tons of online help ([StackOverflow](#) and other forums)



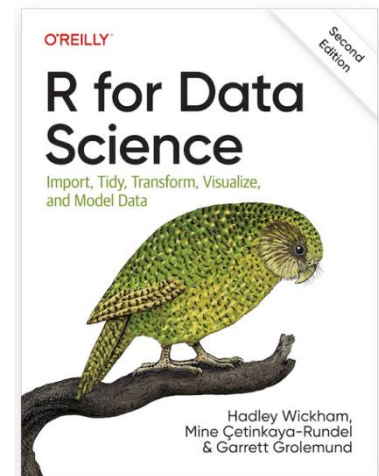
Learning R

- ❑ **Read through the CRAN website:**
lots of links to manuals, help files
- ❑ **QuickR website:** Excellent place to learn
<https://www.statmethods.net>
- ❑ **Learn R in R: Swirl**
<http://swirlstats.com>
- ❑ **Check the Book (tidyverse-focused):**
[R for Data Science](#)



Learn R, in R.

swirl teaches you R programming and data science
interactively, at your own pace, and right in the R
console!



Some helpful links

- [Data Carpentry](#): *Data Analysis and Visualization in R for Ecologists*
- [Tidyverse](#): *R packages for Data Science*
- [Advanced R](#) by Hadley Wickham
- [ggplot2](#): *Elegant graphics for data analysis* by Hadley Wickham
- [Posit Cheatsheets](#) by Posit
- [The R Gallery Book](#) by Kyle W. Brown
- [Introduction to R](#) by Douglas, Roos, Mancini, Cuoto, & Lusseau

Learning R

Five things to remember:

1. R always does exactly what you tell it to do
(But R only tells you what it did if you know how to ask)
2. R assumes that you know what you are saying
(But we often do not know what we are saying)
3. There are a lot of right ways to do everything
(But there are also a lot of wrong ways to do everything)
4. Spaces and capitals are your enemy
5. **BE PATIENT AND DON'T GIVE UP** - the more time you spend using R, the more comfortable you become with it

Assignment (Due Oct 16, 2026)

1. Use the file `Iris_data.csv`

Import the dataset and store it as an object:

- Calculate the mean value of sepal length and sepal width
 - Calculate the Standard Deviation of sepal length and sepal width
 - Make a plot (scatter plot) of sepal length (y-axis) against sepal width (x-axis) and save it as a .pdf file
- * Remember to use what we learned today (as much as you can!)

2. Use the file `Reported_heights.csv`

Import the dataset as an object:

- Calculate the mean value of heights
 - Calculate the standard deviation of heights
 - Make a plot (boxplot) showing the differences between male and female heights and save it as a .pdf file
- * Remember to use what we learned today (as much as you can!)

Once you are done, upload your script (.R – one file) and your plots (.pdf – two files)

Any questions? carolinasarm@usf.edu